SIL PROCUREMENT, SUPPLY CHAIN & INVENTORY

User manuals

All user groups in one volume

User manual · 28 October 2026

Figures, limits and approval chains are the system defaults; your hospital may have changed them.

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User manuals

Which manual is for whom

User guides for SIL Procurement, Supply Chain & Inventory, one per group of users. Everyone should read **Getting started** first. It covers signing in, the menu, notifications and approving documents.

PDF versions for printing and sharing are in `pdf/` : one per manual (`00-index.pdf` … `11-supplier-portal.pdf`) and all of them in one volume (`SIL-user-manuals-complete.pdf`). After changing a manual, regenerate them; see the note at the end of this page.

Manual	For	Built-in role(s)	Sample login*
Getting started	Everyone	–	–
Staff, requesters and heads of department	Anyone who needs goods, services or petty cash; ward and department in-charges	Department / Ward Staff, Department / Ward In-charge (and every other staff role)	`ward.nurse` , `ward.incharge`
Procurement	Procurement officers and managers	Procurement Officer, Procurement Manager	`proc.officer` , `proc.manager`
Stores	Storekeepers	Storekeeper	`storekeeper`
Pharmacy	Pharmacists and pharmacy technicians	Pharmacist, Pharmacy Technician	`pharmacist` , `pharm.tech`
Point of sale	Cashiers at the pharmacy counter (and their supervisors)	POS Cashier	`cashier`
Finance	Finance officers, accountants and the finance manager	Finance / Accounts, Finance Manager	`finance` , `accountant` , `finance.manager`
Petty cash custodian	Whoever holds a petty cash float	Finance / Accounts (custodian of a float)	`accountant`
Supply chain, quality and tender committee	Supply chain manager, quality / cold-chain officer, evaluation committee members	Supply Chain Manager, Quality Officer, Tender Committee	`scm` , `quality` , `tender`
Management and audit	Medical director / management, internal audit	Hospital Management, Auditor	`management` , `auditor`
System administrator	IT / system administrator	System Admin	`admin`
Supplier portal	Suppliers' staff	Supplier	`supplier`

* The sample logins exist only in a training or demonstration installation loaded with the sample hospital. They all use the password set for that installation. In production, users are created by the administrator.

How to read these manuals

- **Menu paths** are written as *Section → Screen*, e.g. *Payments → Payment vouchers*. These are the headings and entries in the left-hand menu. You only see the entries your role allows.
- **Buttons** are written in *italics*, e.g. *Submit for approval*.
- Amounts are in **KES**, VAT inclusive, unless a document says otherwise.
- Approval limits, tolerances and time limits quoted here are the **defaults**. Your hospital may have changed them. The administrator can tell you the values in force.

Who does what (summary)

Process	Raises	Approves	Carries out
Staff requisition (goods / services)	Any staff member	Head of department; Finance manager above 500,000	Stores (stock items), Procurement (other items)
Purchase requisition, RFQ, LPO / LSO	Procurement officer	Procurement manager, then Supply chain, Finance, Management by value	Procurement
Receiving stock (GRN)	Storekeeper / pharmacist	Inspection committee (2 people)	Stores
Accepting non-stock goods and services	Storekeeper / certifying officer	The requester or their department	–
Supplier invoice	Finance	Finance (3-way match exceptions)	–
Supplier payment voucher	Finance (preparer)	Finance manager; Management above 1,000,000	Finance (releaser, not the preparer or an approver)
Petty cash request / advance	Any staff member	Head of department → Finance → Finance manager → Medical director, by amount	Float custodian
Internal (ward) requisition	Ward staff	Ward in-charge of that ward	Store / pharmacy

The system enforces **segregation of duties**: nobody approves their own request, each approval step is by a different person, and the person who prepares a payment cannot release it. A blocked attempt is refused and recorded in the audit log.

Regenerating the PDFs

The PDFs are made from these Markdown files with headless Chrome (the same engine the application uses for its PDFs): a cover page, a table of contents, page numbers and the screenshots embedded. After editing a manual, run `pnpm docs:pdf` (needs Chrome or Chromium; set `CHROMIUM_PATH` if it is not Playwright's own) and commit the PDFs with the change, so the PDFs never drift from the text.

Getting started (all users)

For every user

1. Signing in

1. Open the address your IT team gave you (for example `https://procure.yourhospital.org`).
2. Enter your **username** and **password** and press *Sign in*.
3. **First sign-in.** You must choose a new password before you can do anything else. The password policy (length, character types) is shown if your choice is refused.
4. **Two-factor sign-in (2FA).** If you turned it on, enter the 6-digit code from your authenticator app.

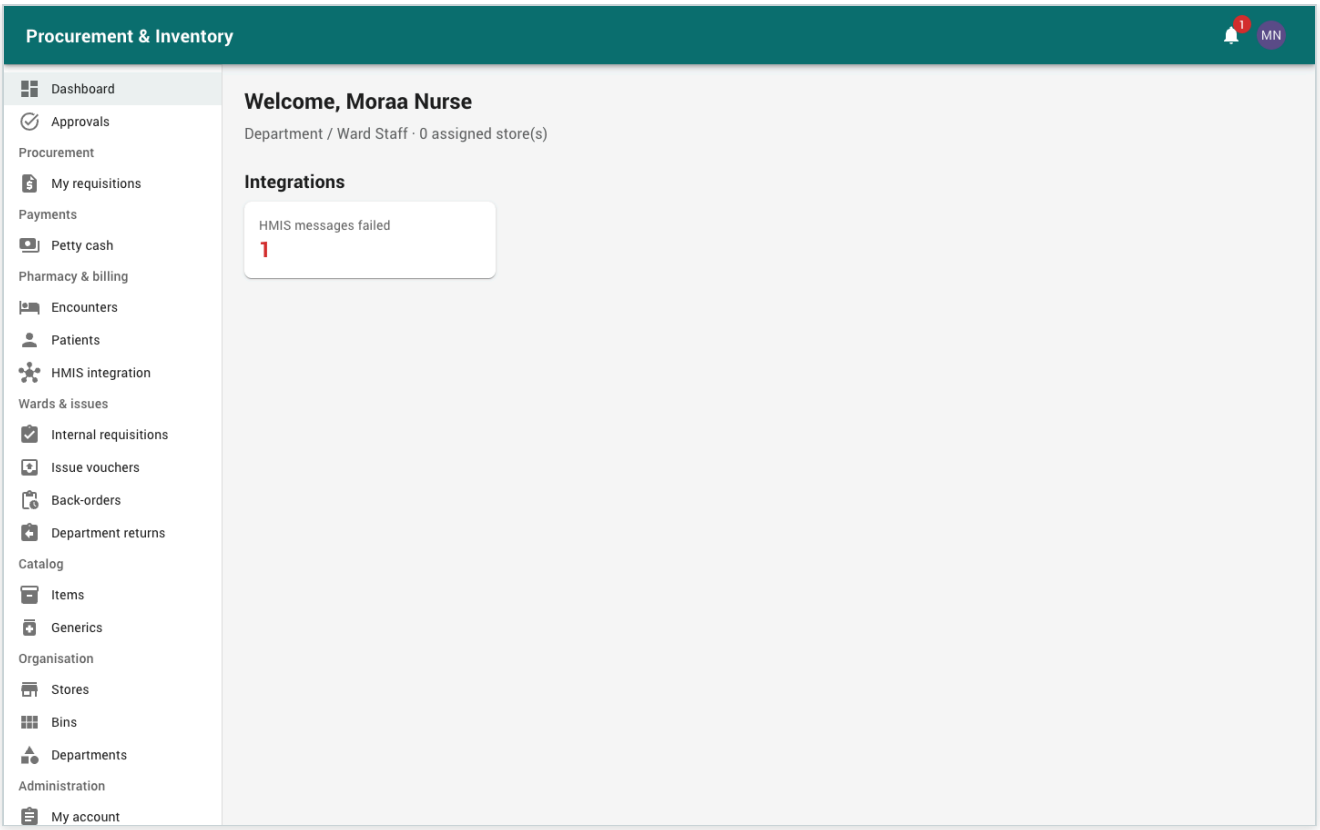
Things to know:

Message	Meaning	What to do
Invalid username or password	Wrong username or password	Try again carefully; usernames are lower case
Account temporarily locked. Try again later.	5 wrong passwords **or** 5 wrong 2FA codes	Wait 15 minutes, or ask the administrator
Too many invalid codes; sign in again	5 wrong 2FA codes on one sign-in	Enter your password again, then a fresh code
Too many requests	Too many sign-in attempts from your network in a minute	Wait a minute
You are signed out while working	30 minutes without activity, or 12 hours since sign-in	Sign in again; unsaved forms are lost

2. Your account (*Administration → My account*)

- **Change password**: enter the current password and the new one twice.
- **Two-factor authentication**: *Set up two-factor authentication* shows a QR code. Scan it with an authenticator app (Google Authenticator, Microsoft Authenticator, Authy), then enter the code shown to confirm. *Turn off* needs your password and a current code. **Finance, procurement and administrator users should turn it on.**
- **Notification preferences**: for each category (Approvals, Requisitions, Payments, Petty cash, stock alerts…) choose *In app*, *Email* or both.

3. The screen



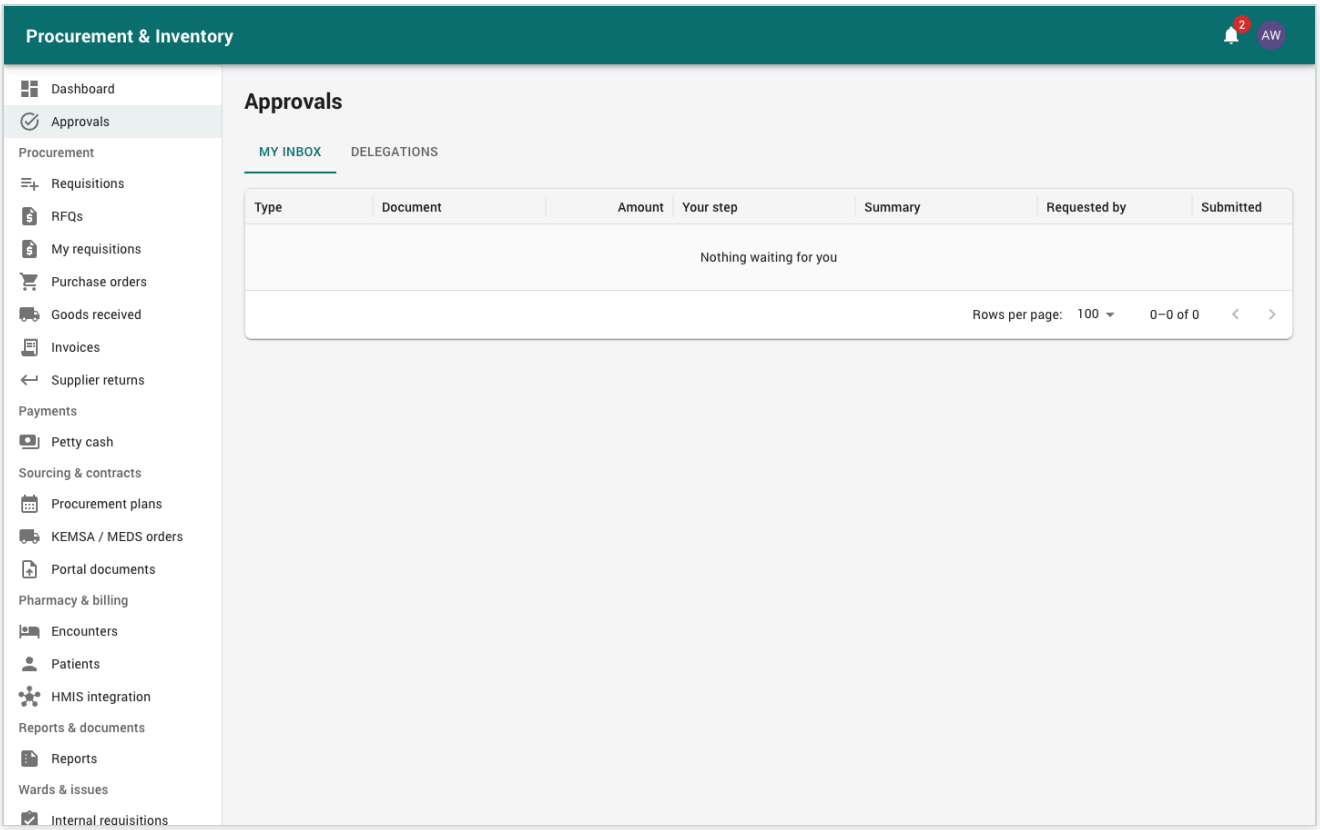
The home page and the left-hand menu (a ward nurse sees only what their role allows)

- **Left menu**, grouped by area (Procurement, Payments, Inventory…). You only see what your role allows. If you need a screen you can't see, ask the administrator to add the right role.
- **Dashboard** (home page) shows tiles for your work: approvals waiting, stock alerts, orders, invoices to pay, petty cash. Click a tile to open the list behind it.
- **Bell** (top right) shows notifications. It refreshes every 30 seconds.
- **Keyboard shortcuts**: `/` jumps to the search box of the page, `Alt+N` opens the *New* dialog, `Enter` submits a dialog and `Esc` closes it.
- **Document chain**: every document page (requisition, order, GRN, invoice, voucher…) shows the documents linked to it, before and after. Click any to open it.
- **PDF**: most documents have a *PDF* button. It prints them with their approval signatures.
- **Attachments**: drag files (PDF, JPEG, PNG) onto a document. Uploading a file with the same name adds a new version; old versions stay available.

4. Approving documents

Every document that needs approval (requisitions, orders, invoices, payment vouchers, petty cash, stock adjustments, disposals…) goes through the same approval engine.

In the app (*Approvals → My inbox*)



Approvals: My inbox, Delegations and Workflows

1. Open **Approvals**. *My inbox* lists everything waiting for you. Each row shows the document, amount, requester, the step and how long it has waited.
2. Open a document to check it. The chain, attachments and history are on the page.
3. Choose:
    - *Approve*;
    - *Reject* (a comment is required; the document stops);
    - *Return for amendment* (a comment is required; it goes back to the requester to change and resubmit).

From an email

Approval emails carry a personal link. It opens a page showing the document summary with *Approve*, *Reject* and *Return for amendment*, without signing in.

- The link works **once**, only for you and only for that step.
- It expires after **72 hours**. After that, use the app.
- If someone else has already acted on the step, the page says the approval *has moved on*.

Rules you will meet

- You can never approve something you requested or prepared.
- Each approval in a cycle must be by a different person.
- If you don't act in time, you may get a reminder, and the step may be **escalated** to another role, whose members can then act on it too.

Delegation (*Approvals → Delegations*)

Going on leave? Create a delegation: *Delegate to* a colleague, from and to dates, and a reason. During that time they can approve **on your behalf**; the history records "on behalf of". End it early from the same tab.

5. Statuses you will see

Status	Meaning
Draft	Being prepared; only the author sees and edits it
Pending approval / Submitted	In the approval chain
Approved	All approval steps done
Returned	Sent back for changes; edit and resubmit
Rejected	Stopped by an approver
Cancelled	Withdrawn
Closed	Finished

6. Getting help

- Something refused with a message in capitals (e.g. `OVER_LIMIT`, `ACCEPTANCE_REQUIRED`): the manual for your role explains the common ones.
- For access problems contact the **system administrator**. For approval limits and workflows, contact the **supply chain manager** or **finance manager**.

Staff, requesters and heads of department

For all staff and department in-charges

For every staff member who needs goods, services or cash, and for ward / department in-charges (heads of department) who approve their staff's requests.

You are	Built-in role	What you do here
Any staff member	any staff role (e.g. Department / Ward Staff)	Raise requisitions and petty cash requests, accept deliveries, collect goods
Ward / department in-charge (HOD)	Department / Ward In-charge	The same, plus approve your department's requests, sign off service deliveries and department plans

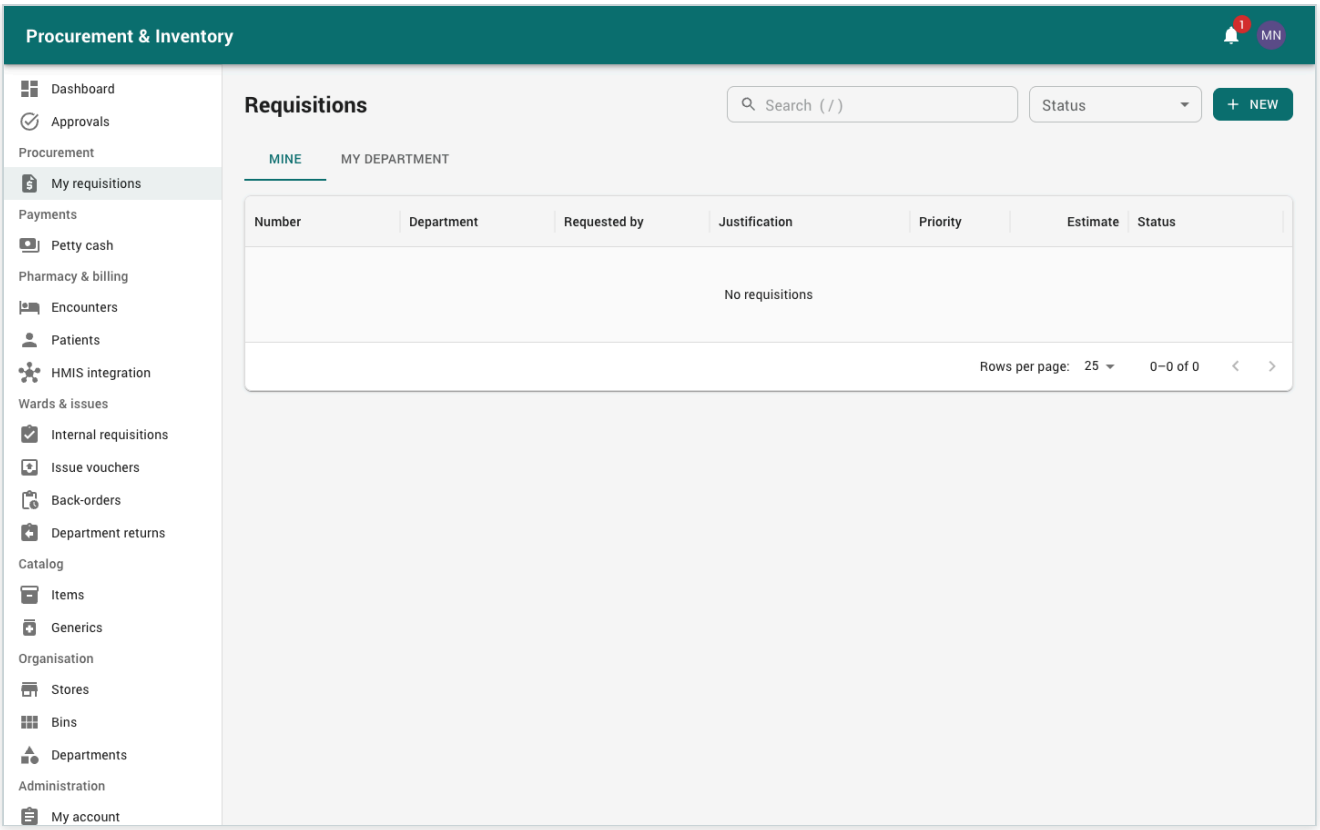
Read Getting started first.

1. Which request do I use?

You need	Use	Menu
Anything for your department: stock items, goods the hospital does not stock, a service or repair	**Staff requisition**	*Procurement → My requisitions*
Routine ward stock from the store or pharmacy (top-ups, consumables, drugs)	**Internal requisition**	*Wards & issues → Internal requisitions*
A small cash payment, or an advance you will account for	**Petty cash request**	*Payments → Petty cash*
Record ward stock used on an admitted patient (so it is charged)	**Charge ward consumables**	*Pharmacy & billing → Encounters*

A staff requisition may mix stock items and non-stock items. Stock items are sent to the store automatically; everything else goes to Procurement.

2. Staff requisitions (*Procurement → My requisitions*)



My requisitions: Mine / My department, with New at the top right

Raising one

1. *My requisitions* → *New* (or `Alt+N`).
2. Fill in the header:
   - **Department / cost centre**: your own department, or another one if you raise it on their behalf;
   - **Project** (optional), **Priority**, **Required by**;
   - **Justification**: why it is needed. Approvers read this;
   - **Store for the stock items**: the store that should issue any stock lines;
   - **Budget** line (optional).
3. Add lines with *Add line*:
   - **Stock item**: pick the catalog item and quantity;
   - **Goods / service not in stock**: describe *What is needed*, the unit, quantity and **estimated cost**; add *Notes / specification*. Pick the **GL account** if you know it (your hospital may make it required).
4. Attach quotes, specifications or photos.
5. *Save draft* to finish later, or *Submit for approval*.

When you submit, the **budget** is checked. Depending on the budget's setting you either see a warning, or the submission is refused until the budget is increased.

Approval

Amount	Approvers (default)
Up to 500,000	Head of department (in-charge of the requesting department)

Amount	Approvers (default)
Above 500,000	Head of department → Finance manager

You are notified at each step. If it is **Returned**, open it, make the changes asked for in the comment, and submit again.

Following it

The requisition page shows **Where it is** for every line:

- **Stock lines** become an approved internal requisition at the store. Stock is reserved for you and the store issues it.
- **Other lines** go to the Procurement queue, then onto a purchase request, an order (LPO / LSO), delivery and hand-over to you.

Statuses: Draft → Submitted → Approved → Partly converted / Converted → **Closed** (when everything is delivered). Also Returned, Rejected, Cancelled.

You get a notification when:

- the purchase request is approved;
- the order (LPO / LSO) is sent to the supplier;
- a delivery is **waiting for your acceptance**;
- goods have been received from the supplier;
- goods are **ready for collection** (with a one-time code if they hand over at the counter);
- the requisition is delivered and closed.

Cancel requisition (with a reason) withdraws it while nothing has been ordered. *PDF* prints it with the approval signatures.

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3. Accepting a delivered service or non-stock goods

When a supplier delivers a service (repair, maintenance, consultancy) or goods bought for your requisition, the receiving officer records it. **You, or your department's in-charge, must then accept it** before the supplier can be paid.

1. Open *My requisitions*. The **Waiting for you** panel lists deliveries to accept. (In-charges also see them under *Logistics → Service certificates*.)
2. Open the certificate. Check the quantities accepted and rejected, the notes and the attachments (job card, signed note, photos).
3. Choose *Accept delivery*, or *Reject* with a reason (the officer must correct it).

What acceptance does:

- the order counts as received, so the supplier's invoice can be matched and paid;
- a **service** accepted by you counts as delivered to you;
- **goods** still have to be handed over to you (next section).

Stock levels never change for non-stock items.

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4. Collecting goods (hand-over)

When received goods are ready, Procurement or Stores hand them over to you or a colleague in your department. You are told they are **ready for collection**.

You confirm receipt in one of two ways:

- **In the app**: open the hand-over (from the notification, or the *Waiting for you* panel on *My requisitions*) and click *I have received these*.
- **At the counter**: the officer clicks *Send code to the receiver* and you get a **6-digit code**. Tell it to the officer. The code is valid for 15 minutes; after 5 wrong attempts it stops working and a new one must be sent.

When every line is delivered, the requisition **closes** and you are told.

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5. Internal (ward) requisitions (*Wards & issues → Internal requisitions*)

For routine stock from a store or pharmacy to your ward or department.

1. *New*: choose the **store** to supply you, priority and required-by date.
2. Add items and quantities (in base units, e.g. tablets, pieces). For an admitted patient, add the patient / encounter on the line, so the item is charged to them.
3. *Submit*. Your **ward in-charge** approves. Only the in-charge of *your* department can approve, never you.
4. On approval the stock is **reserved** for you. The store issues it oldest-expiry-first (FEFO) and prints an issue voucher.
5. If the store doesn't have everything, the rest stays on **back-order** and is issued when stock arrives.

Statuses: draft → pending approval → approved → partly issued (back-order) → issued, or closed if a back-order is cancelled.

Returning stock (*Wards & issues → Department returns*): choose the issued lines, quantity, reason and condition (resaleable or damaged / expired). The store posts the return and your department is credited.

Consumption (*Wards & issues → Consumption*, in-charges): what your department used, by period, item or patient, at cost.

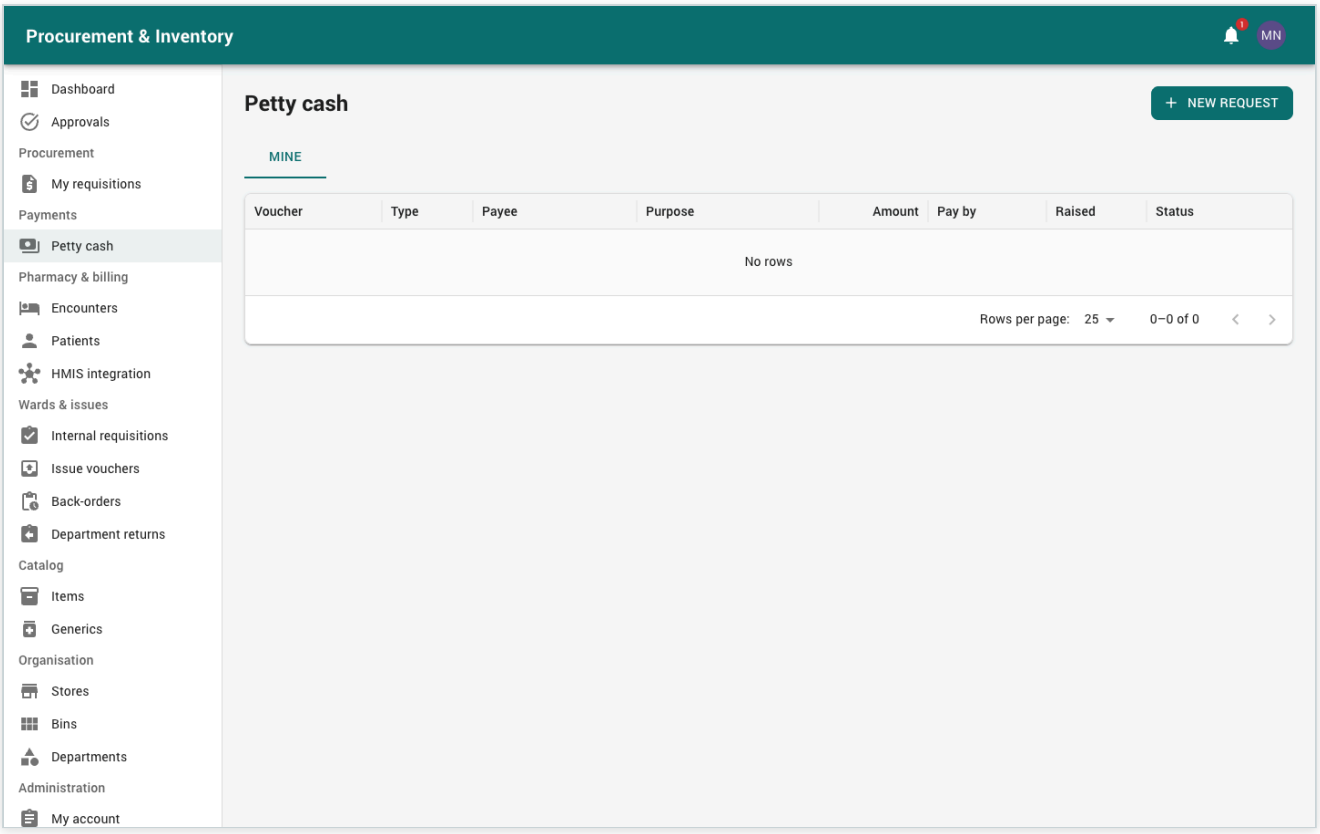
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6. Charging ward stock to a patient (*Pharmacy & billing → Encounters*)

Ward or theatre stock already issued to your ward and used on an admitted patient: open the patient's **encounter** → *Charge ward consumables* → items and quantities. No stock moves (it was issued to the ward already); the patient's bill is charged once.

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7. Petty cash (*Payments → Petty cash*)



Petty cash requests

Request a payment or an advance

1. *New*: choose the **float** (e.g. your department's or Administration).
2. Type:
    - **Payment**: to a staff member or a vendor;
    - **Advance / imprest**: cash to you, which you must account for later.
3. Fill in the **purpose**, **expense account**, **cost centre**, optional project and the **amount**. Attach receipts or quotes.
4. **Pay by**: cash, **M-PESA to a phone**, **PayBill + account** or **Till**. M-PESA is only possible on floats that pay by M-PESA, and in whole shillings.
5. *Submit for approval*.

Checks you may meet:

Message	Meaning
`OVER_LIMIT`	Above the float's per-transaction limit; use a requisition instead
`INSUFFICIENT_FLOAT`	The float hasn't enough available cash now; try later or another float
`POSSIBLE_DUPLICATE`	Same payee and amount within 7 days. If it is genuinely different, fill in *Why this is not a duplicate*; approvers will see the warning
`ADVANCE_OUTSTANDING`	You have an advance not yet retired; retire it first

Approval (default)

Amount	Approvers
Up to 10,000	Your head of department → Finance
10,000 – 50,000	+ Finance manager
Over 50,000	+ Medical director / CEO

Getting paid

Once fully approved, the **float custodian** pays you:

- **cash**: collect it, then confirm in the app with *I received it*;
- **M-PESA**: the money arrives on your phone; the request is marked paid when M-PESA confirms.

Retiring an advance

Within **7 days** of receiving an advance:

1. Open the advance → *Retire the advance*.
2. *Add receipt* for each expense: amount, expense account, receipt reference, and attach the receipt image.
3. Submit. The custodian checks the receipts and either accepts them or returns them to you for correction.
    - Spent **less**: return the change to the custodian; it goes back into the float.
    - Spent **more**: the custodian pays you the difference.

You get daily reminders when a retirement is overdue, and you can't take a new advance until it is retired.

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8. For heads of department (in-charges)

Approving

Approvals → *My inbox*. You see your department's:

- staff requisitions;
- internal requisitions;
- petty cash requests;

- purchase plans.

Check the justification, quantities, amounts and attachments, then *Approve*, *Reject* or *Return for amendment*. See Getting started → Approving, including approving by email and **delegating** while you're away.

You can't approve a request you raised yourself; it goes to another approver in the chain.

Signing off deliveries

As in-charge you can accept or reject service and non-stock deliveries for your department (section 3), even when the requester is away.

Department plan (*Sourcing & contracts → Procurement plans*)

Prepare your department's annual procurement plan: items, quantities, quarter and budget vote head. Submit it for approval (procurement → finance → management). Mid-year revisions create a new version through approval; the approved version stays in force meanwhile.

Watching spend

Reports (if your role allows): consumption by department, pending requisitions, delivered vs outstanding.

Procurement officers and managers

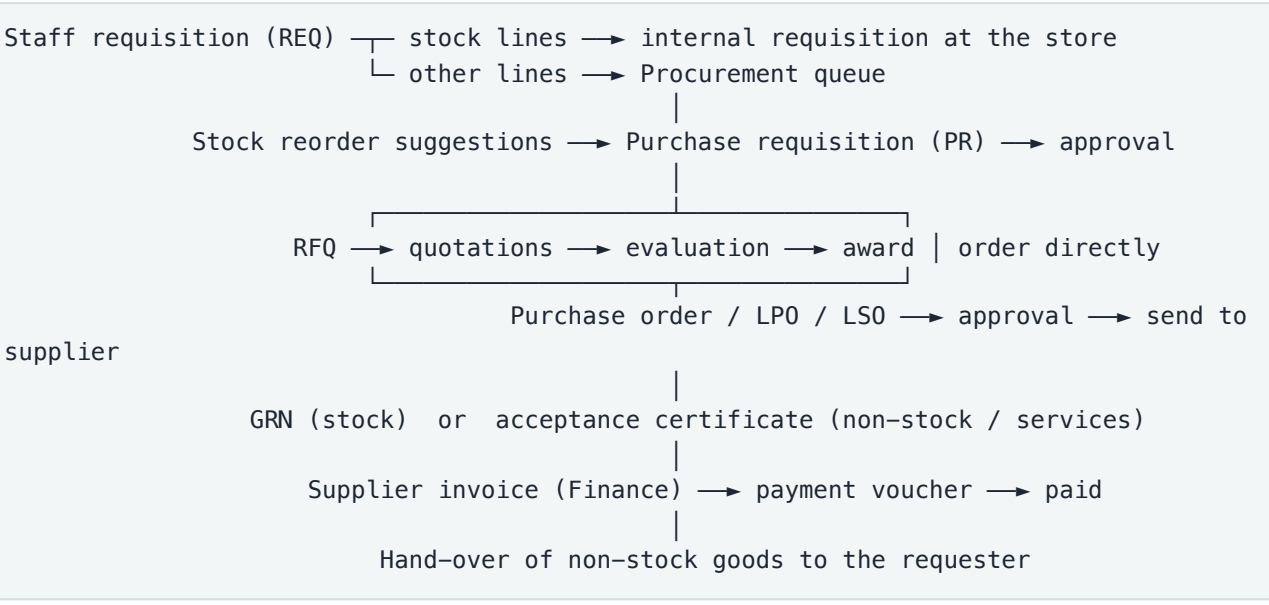
For procurement officers and managers

For the **Procurement Officer** (buys: purchase requisitions, RFQs, orders, suppliers) and the **Procurement Manager** (approves purchase requisitions and orders, manages tenders and contracts). Read Getting started first.

Your menu: *Procurement* (Requisitions, RFQs, My requisitions, Procurement queue, Purchase orders, Goods received, Invoices, Suppliers, Budgets, Supplier returns), *Sourcing & contracts* (plans, tenders, contracts, supplier performance, KEMSA / MEDS orders, KPIs), *Logistics* (Inbound shipments, Service certificates), *Reports*.

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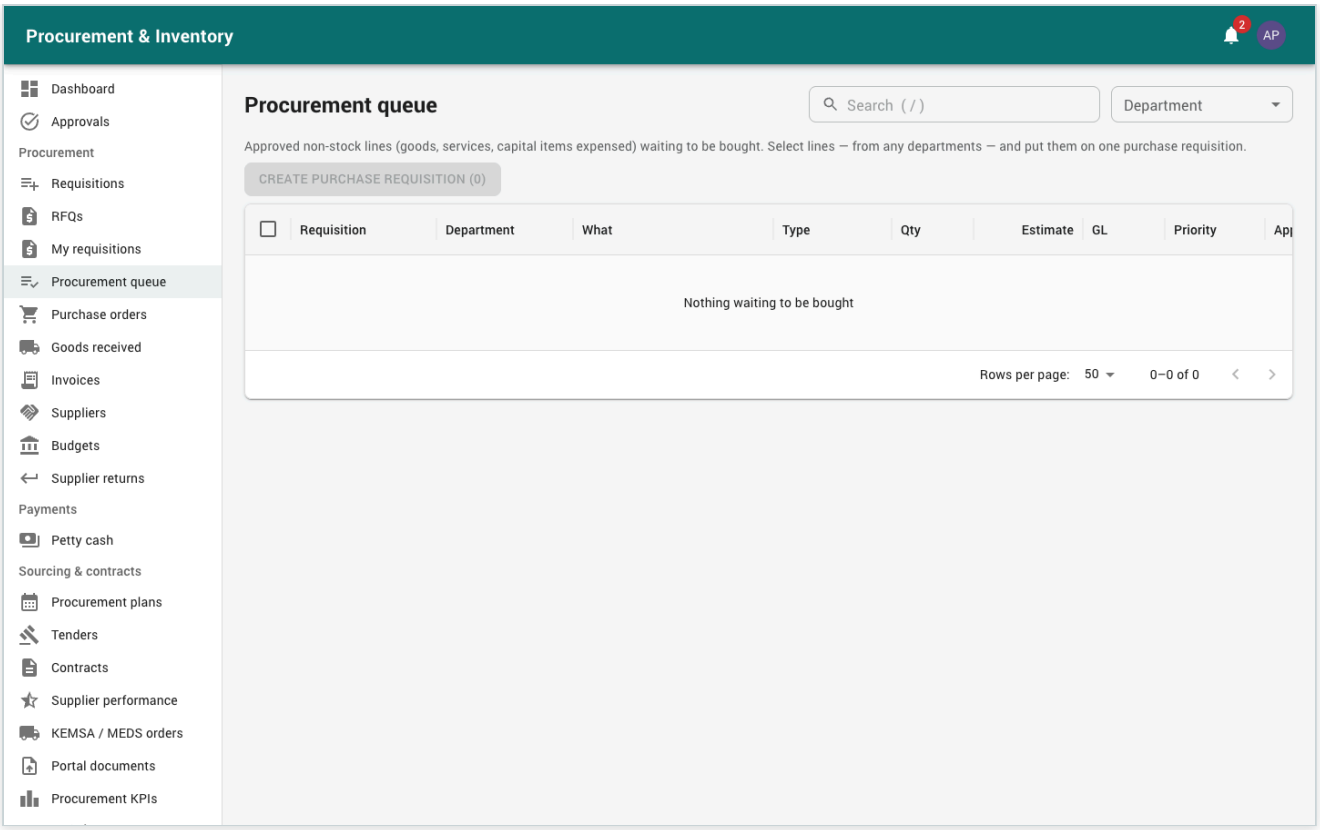
1. The buying cycle



Every document is linked; the **document chain** on each page shows the whole path.

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2. Procurement queue → purchase requisition



The procurement queue: approved non-stock lines from every department

Procurement → Procurement queue lists **approved non-stock lines** (goods, services, capital items expensed) from every department's staff requisitions.

1. Tick lines, from one or several departments, that you will buy together.
2. *Create purchase requisition (n)*. Fill in:
    - **Owning department** and **Delivery location**;
    - **Procurement method** (Direct, RFQ, tender…). *Recommended by value* shows the method the value calls for; choosing another needs a *Method justification*;
    - **Suggested suppliers** and a justification.
3. The purchase requisition (PR) is created as a **draft**. Open it under *Procurement → Requisitions* and *Submit for approval*.

If the PR is rejected or cancelled, its lines go back to the queue.

Stock items reach you differently: from *Planning → Suggested reorder* or *Demand forecasting* (selected suggestions become a draft PR), or a PR raised directly under *Requisitions → New* (items in base units, justification, priority, required-by date, budget).

PR approval (default)

Value	Approvers
< 100,000	Procurement manager
100,000 – 1,000,000	Procurement manager → Supply chain manager
≥ 1,000,000	+ Hospital management

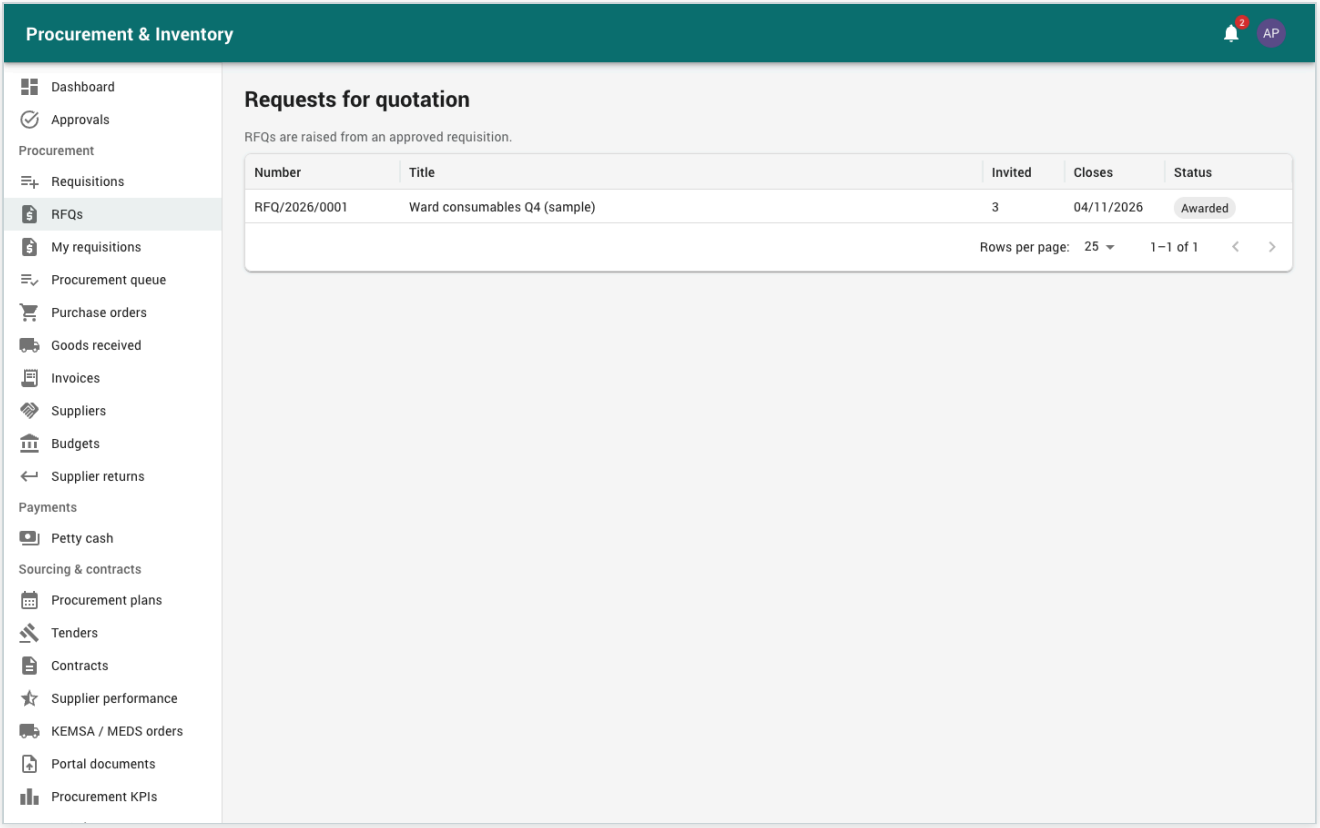
Value	Approvers
Contains controlled-drug items	Pharmacist → Procurement manager

The sourcing method is checked on submission against the value bands: low value < 50,000; RFQ 50,000–1,000,000 with 3 bidders; open tender ≥ 1,000,000; restricted tender, RFP and direct procurement need a justification.

PR statuses: Draft → Pending approval → Approved → **RFQ issued** → **Partially fulfilled** → **LPO raised** / Closed. Cancelling or expiring an order gives its quantities back to the PR.

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3. Requests for quotation (*Procurement → RFQs*)



Requests for quotation

Raising and sending

1. From an **approved PR**: *Request quotations* → title, closing date and *Invite suppliers*. Only **active** suppliers with valid prequalification can be invited.
2. *Send invitations*. Each supplier gets an email with the RFQ PDF; suppliers on the portal also see it there.
    - At least **3 suppliers** must be invited. With fewer, you must give a justification, otherwise it is refused (`MIN_SUPPLIERS`).

Quotations

Record a quotation for each supplier (or the supplier submits it in the portal):

- unit price per line, in any currency with an exchange rate (blank = KES);
- delivery days, payment terms, compliant or not (*Why not compliant*).

The **Comparative analysis** converts every price to KES at today's rate, ranks the compliant quotes and highlights the lowest. When the closing date has passed, *Close bidding*.

Evaluation (for quality-and-cost selection)

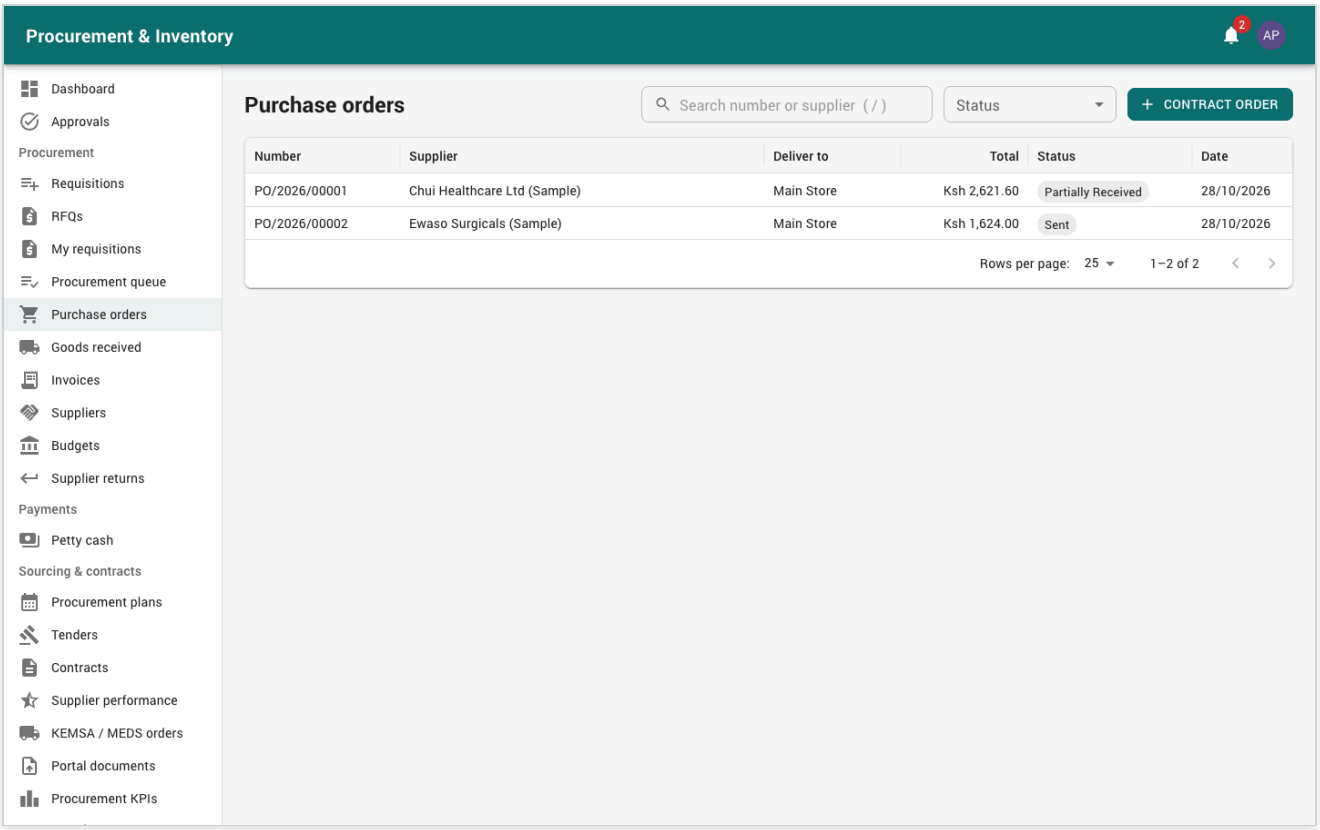
1. *Evaluation set-up* (before scoring): **technical weight** (0–100 %), **technical pass mark** (default 70) and the **evaluation committee**. A weight of 0 means price only; no evaluation is needed.
2. Each committee member opens the RFQ and enters *Your technical scores* (0–100 with a comment) for every compliant quotation.
3. When all scores are in: *Complete the evaluation*, with the committee recommendation.
    - Financial score = lowest compliant total ÷ this total × 100.
    - Combined = technical × weight + financial × (100 − weight).
    - A quotation below the pass mark cannot be recommended.

Award

1. In the comparative sheet, the recommended (or lowest compliant) supplier is pre-selected for each line. Change a choice if needed.
2. Awarding a line to anyone other than the lowest compliant quote needs an **Award justification** (`JUSTIFICATION_REQUIRED`).
3. Set **Expected delivery** and, for non-stock, the **Delivery address / site**.
4. *Award and create orders*: one **draft order per supplier**, in the quotation's currency with the exchange rate fixed.

---

4. Purchase orders, LPOs and LSOs (*Procurement → Purchase orders*)



Purchase orders

Orders are created from an RFQ award, from an approved PR (*Order directly (no RFQ)*), or with *Order from contract price list*. An order with only non-stock lines is a **Local Purchase Order (LPO)**. If every line is a service it is a **Local Service Order (LSO)**.

On the order:

- lines in an ordering unit (e.g. BOX of 100) converted to base units; VAT per line;
- non-stock lines: description, unit (JOB for services, PCS otherwise), GL account;
- **withholding tax** (default 5 % on services) on the pre-VAT amount, which you can override per line. The order shows the WHT and the **Net payable**;
- *Valid until*: order date + 60 days; *Terms*; *Deliver to*.

Approval (default)

Value (KES)	Approvers
< 500,000	Procurement manager
500,000 – 5,000,000	Procurement manager → Finance
≥ 5,000,000	+ Hospital management

Approval **commits the budget**. A BLOCK budget refuses approval when exceeded.

Sending

Send to supplier emails the order with its PDF (some hospitals send automatically on approval). Suppliers on the portal see it and acknowledge it there.

Changing an order

- *Amend* (with *Reason for the amendment*): creates a new **version** that needs re-approval on the new value. *Version history* keeps every version. A rejected amendment restores the previous version.
- *Cancel purchase order* (with reason): only if nothing has been received. The budget commitment is released and the quantities go back to the PR.
- *Close purchase order*: stop waiting for the rest; releases what is still committed.
- **Expiry**: approved or sent orders past *Valid until* with nothing received or invoiced expire overnight. The commitment is released, the quantities return to the PR, and you are notified.

Order statuses: Draft → Pending approval → Approved → Sent → Partially received → Received → **Invoiced** → **Paid**; also Cancelled, Closed, Expired.

---

5. Receiving

- **Stock items** are received by Stores on a **GRN** (see Stores).
- **Non-stock goods and services** are recorded on an **acceptance certificate** (*Logistics → Service certificates → Record delivery / completed work*). A storekeeper or certifying officer records accepted and rejected quantities. **You can't record it for an order you raised or approved.** When the order was bought for staff requisitions, the requester or their in-charge must *Accept delivery* before it can be posted (`ACCEPTANCE_REQUIRED`).
- **Inbound shipments** (*Logistics → Inbound shipments*): track shipped and expected dates, carrier and tracking for important orders. Delays are flagged daily.

6. Hand-over to the requester

On the staff requisition page, *Delivery to the requester* shows each non-stock line as ordered / received / delivered / waiting.

1. *Hand over*: choose the receiver (the requester or an active colleague in their department), the quantities received and not yet handed over, the condition and notes.
2. The receiver confirms in the app (*I have received these*), or at your counter: *Send code to the receiver* and type the 6-digit code they give you (15 minutes, 5 attempts).
3. When everything is delivered the requisition closes automatically.

You can only hand over what has been received (`NOT_RECEIVED`).

---

7. Suppliers (*Procurement → Suppliers*)

- Register: name, **KRA PIN** (validated, e.g. `P051234567X`), contacts, categories supplied, **prequalification** status and expiry, payment details (the bank account is stored encrypted and shown masked).
- Attach licences and certificates with their expiry dates. *Reports & documents → Documents → Expiring supplier documents* lists those expiring soon.

- Status: active / suspended / blacklisted (with reason). Suspension, blacklisting and reinstatement go through approval (supply chain → management). Only **active, prequalified** suppliers can be invited or ordered from.
- **Performance** (*Sourcing & contracts → Supplier performance*): on-time delivery, quality rejects, price variance, and monthly scorecards. A score below 60 opens a corrective action.
- Suppliers with portal users quote, bid, acknowledge orders and submit invoices online (see Supplier portal).

8. Budgets (*Procurement → Budgets*)

Per department × vote head × financial year. A PO approval **commits**; an approved invoice turns the commitment into **actual**; cancelling or closing **releases** it. Each budget is WARN (warning shown) or BLOCK (refused when exceeded). Report: *Budget vs committed vs actual vs paid*.

9. Tenders, contracts and plans (Procurement manager)

- **Annual procurement plans** (*Sourcing & contracts → Procurement plans*): departments prepare them; you **consolidate** the approved plans into the hospital plan, by item and vote head.
- **Tenders**: notice → bid documents → clarifications and addenda → sealed bids (portal or bid register; no prices are visible before opening) → **opening** with attendees and minutes → **evaluation committee** (one chair; each member signs a conflict-of-interest declaration; preliminary, technical and financial evaluation) → recommendation → professional opinion → award approval (supply chain → management) → award and regret letters → contract. Scores are locked once the evaluation is locked.
- **Contracts and framework agreements**: prices, quantities, validity, performance securities and milestones. **Call-off orders** become draft POs at contract prices within the remaining balance. **Variations** apply after approval (procurement → finance). Alerts at 90 / 30 days before expiry.
- **KEMSA / MEDS orders**: order templates per store and cycle raise draft POs when due.

10. Approving (Procurement manager)

Approvals → My inbox: purchase requisitions and orders in your band. Check value, supplier, budget line and justification (and, for awards, the comparative sheet and the award justification). You can't approve a document you raised.

11. Reports you'll use

On *Reports*: Pending requisitions · Pending purchase requests · Open orders / LPOs · Outstanding POs · RFQ status and awards · GRN / acceptance registers · Received not invoiced · Three-way match exceptions · Purchases over time · Supplier performance · Price history · Procurement cycle time · Budget vs actual · Delivered to users vs outstanding · Procurement KPIs. Each one can be exported (CSV / Excel / PDF) or *emailed on a schedule*.

12. Common messages

Message	Meaning
`MIN_SUPPLIERS`	Fewer than 3 suppliers invited without a justification
`JUSTIFICATION_REQUIRED`	Award past the lowest compliant quote without a justification
Complete the technical evaluation before awarding	The RFQ has a technical weight and isn't evaluated yet
Supplier not eligible	Supplier suspended, blacklisted or prequalification expired
Budget exceeded	BLOCK budget: ask Finance to raise it, or reduce the order
`ACCEPTANCE_REQUIRED`	The requester must accept the delivery first
`NOT_RECEIVED`	Hand-over of more than has been received
`NON_STOCK_LINE`	A GRN can't receive non-stock lines; use an acceptance certificate

Stores (storekeepers)

For storekeepers

For storekeepers of the main store and satellite stores. Pharmacists who run a pharmacy store use the same screens for receiving, transfers, adjustments and stock takes. Read Getting started first.

Your menu: *Inventory* (Stock on hand, Receipts, Transfers, Adjustments, Stock takes), *Wards & issues* (Internal requisitions, Issue vouchers, Back-orders, Department returns), *Procurement* (Goods received, Supplier returns), *Planning* (Suggested reorder, Near expiry, Stock levels, Stock alerts), *Logistics* (Distribution, Service certificates, Labels).

You only see and move stock in **the stores assigned to you**.

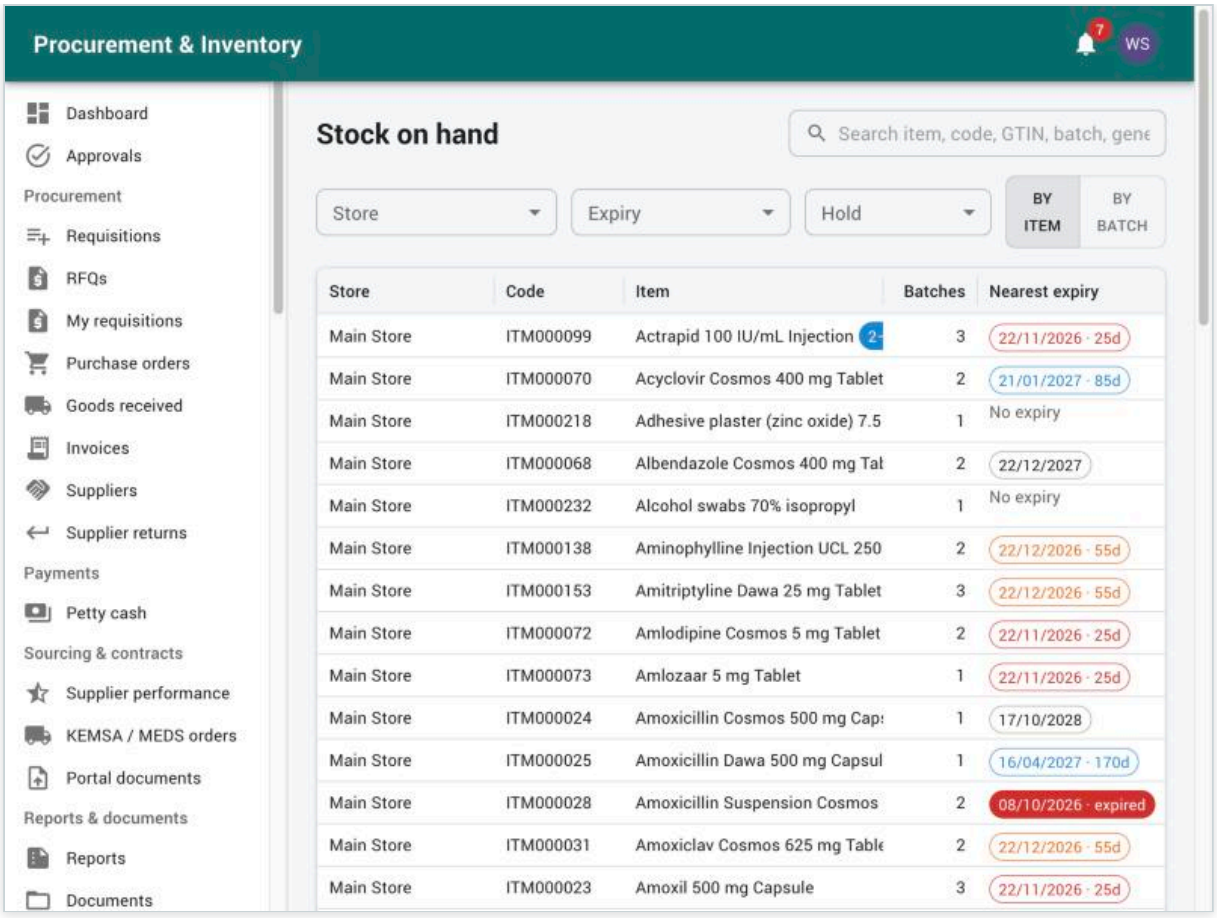
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1. How stock works (the rules the system enforces)

- All stock is kept in **base units** (tablet, piece, ml). Packs are converted.
- Every movement is written to the **stock ledger** and cannot be edited or deleted. A mistake is corrected by a new movement (adjustment or return).
- Stock leaves by **FEFO** (first expiry, first out). Picking another batch needs the *override FEFO* permission and a reason, which is recorded.
- The system refuses:
  - negative stock;
  - issuing **expired** batches, batches **on hold** (quarantine, recall, expired) or stock in the quarantine bin;
  - receiving goods that are already expired;
  - any movement dated on or before the store's **period lock** date;
  - movements of items frozen by a **stock take** in progress;
  - **controlled drugs** without a second signatory (witness username and password) who holds the controlled-drug permission;
  - cold-chain or controlled items moved by someone without the matching permission.
- **Available** = on hand − reserved. Stock reserved for approved requisitions can't be issued to others, transferred or returned to a supplier.

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2. Stock on hand (*Inventory → Stock on hand*)

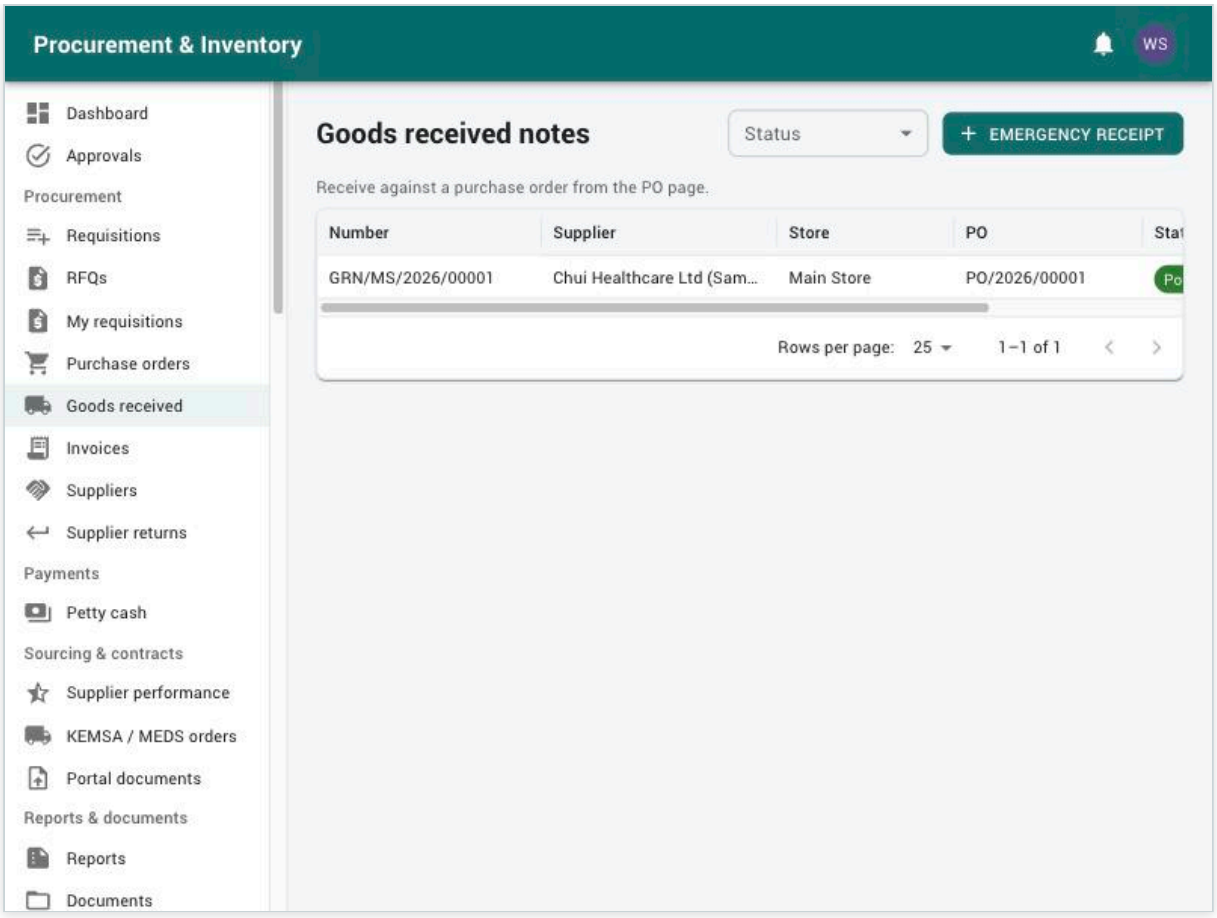


Stock on hand by item, with the nearest expiry

- Filter by store, bin, item, generic, category, batch; *Group by* item or batch; show *Expired* or held stock; search by name or code.
- **Bin card**: open an item to see every movement with a running balance, where it came from or went to, the patient and the witness.
- **Batch trace**: every movement of a batch, where it is now and which patients got it.

---

3. Receiving from suppliers: GRN (*Procurement → Goods received*)



Goods received notes: receive from the purchase order; Emergency receipt for goods without one

1. Open the **purchase order** (*Procurement → Purchase orders*; only approved / sent orders with something outstanding) → *Receive goods*. If the order arrived on a notified shipment, pick the shipment. Enter the **Delivery note** number.
2. For each line, enter the quantity received in the **delivery unit** (converted to base units), the **batch**, manufacture and **expiry** dates (required for batch-tracked items) and the **bin**. You can *Scan barcode* to find the line. For cold-chain items, enter the *Arrival temperature*: outside 2–8 °C puts the batch on hold.
3. Attach the delivery note and certificates of analysis.
4. *Send for inspection*.
5. **Inspection committee**: at least **2 people**, and not the receiver, open the GRN, *Record results* (accepted and rejected quantities with reasons) and *Confirm as committee member*.
6. *Post accepted goods to stock*.

What posting does:

- accepted quantities enter stock at the order cost (VAT included);
- rejected quantities create a **return to supplier** automatically;
- the order's received quantities are updated;
- back-orders waiting for these items are re-reserved.

Rules:

- more than ordered (beyond the tolerance, default 0 %) is refused;
- the receiver can't be the person who requested the PR or approved the order;

- **Non-stock lines** (services, items not kept in stock) can't be received on a GRN (`NON_STOCK_LINE`); see section 4.

Emergency receipt (*Goods received → Emergency receipt*; no order yet, `purchasing.emergency_receipt`): receive without a PO, with *Why there is no PO*. It stays flagged until you *Regularise with PO*.

---

4. Receiving non-stock goods and services (*Logistics → Service certificates*)

For repairs, services and goods not held in stock:

1. *Record delivery / completed work* → choose the order.
2. For each line: **accepted** quantity, and any **rejected** quantity with its reason. Partial deliveries are fine.
3. Add inspection notes and the signed job card, delivery note or photos.
4. If the order was for a staff requisition, the **requester or their in-charge** must *Accept delivery* before you can post (`ACCEPTANCE_REQUIRED`). Otherwise post it yourself.

You can't record a delivery for an order you raised or approved. **Stock levels never change** for these lines.

Then **hand the goods over** to the requester. On the requisition: *Hand over* → the receiver confirms in the app or with the **6-digit code** they get (*Send code to the receiver*). See Procurement → Hand-over.

---

5. Issuing to wards and departments

Internal requisitions (*Wards & issues → Internal requisitions*)

Wards raise them; their in-charge approves; on approval the stock is **reserved** for them. You see approved requisitions for your store waiting to be issued (also on the dashboard).

Issue voucher

1. Open the requisition → *Issue*. The **issue plan** proposes FEFO picks (batch, bin, expiry) for everything outstanding.
2. Confirm each pick, ideally by scanning the item barcode or batch number (*Scan item barcode or batch number*). A code that doesn't match is refused. Your hospital may make scanning compulsory.
3. To pick a different batch, enter the *FEFO override reason* (needs the permission).
4. Controlled drugs: a **witness** enters their username and password.
5. Enter who collected it and post. Print the **issue voucher PDF** (batch, expiry, bin, value, signatures).

Whatever can't be supplied stays on **back-order** (*Wards & issues → Back-orders*). When stock arrives it is reserved automatically, most urgent and oldest first, and you are told which back-orders can now be filled. A back-order can be cancelled with a reason.

Department returns (*Wards & issues → Department returns*)

Post a ward's return: lines from what was issued to them, reason and condition.

- **Resaleable** goes back to the same batch in a usable bin.
- **Damaged / expired** goes to the **QUARANTINE** bin; stock there is never issued.

A department can't return more than it was issued. It is credited at the cost it was charged.

---

6. Transfers between stores (*Inventory → Transfers*)

1. The **receiving** store requests: *Request stock from another store → Request from* (store), items, quantities (base units) → *Send*.
2. The **supplying** store opens it → *Dispatch*. Picks are FEFO; overriding needs a reason. The stock is now **in transit**.
3. The receiving store → *Confirm receipt*, quantities received, *Put away to bin*. A shortfall needs a *Discrepancy reason* and is written off from transit.

Planned routes from the main store to satellites (picking lists, packing, dispatch note, proof of delivery per stop) are under *Logistics → Distribution*.

---

7. Adjustments (*Inventory → Adjustments*)

For breakage, losses, found stock, expiry write-offs.

1. *New stock adjustment* → store, **reason** (each reason is an increase or a decrease), items, batch (or *New / found batch*), quantity, notes and attachments.
2. *Submit for approval*. An approver who is not you approves it; large adjustments need a **second, different approver**.
3. It posts on final approval. A rejected adjustment shows the *Rejection reason*.

Expiry write-offs get a **disposal certificate** PDF once posted.

---

8. Stock takes (*Inventory → Stock takes*)

1. *New stock take*: a *Full count* (the whole store is frozen) or a *Cycle count (selected items)*. Counts can be **blind** (system quantities hidden).
2. *Freeze and start counting*: a snapshot is taken and movements of those items are blocked.
3. Print the *Count sheet*, count, enter the counted quantities and *Save counts* (as many times as needed).

4. *Submit for review*. The **Variance** and value per line are shown.
5. Someone other than you reviews it → *Approve and post variances*. Stock is corrected and the items are unfrozen.

Plan counts out of hours. Nothing can be issued or received for frozen items.

---

9. Returns to suppliers (*Procurement → Supplier returns*)

- **From stock**: supplier, store, reason and batch lines (the GRN and order are found from the batch) → *Dispatch*. Expired, held and quarantined stock may be returned; reserved stock may not.
- **Inspection rejects** from a GRN are created automatically.
- Record the supplier's **credit note** (number, date, amount) when it arrives; any difference is shown.

---

10. Planning and alerts (*Planning*)

- **Stock levels**: per item per store, min / max / reorder level / reorder quantity / safety stock / lead time. **AUTO** levels are recalculated nightly from average monthly consumption (AMC); editing a level switches the row to **MANUAL**.
- **Suggested reorder**: items at or below their reorder level, with suggested quantities. Select, adjust, and press one button:
  - the **main store** (one that buys) gets a draft purchase requisition, sent to Procurement;
  - a **satellite store** gets a transfer request to its parent store.
- **Near expiry**: batches expiring within 30 / 60 / 90 / 180 days, their value, whether they will be used up in time, and the store using the most of them. Actions: **redistribute** (transfer), **return to supplier**, **write off** (adjustment with reason EXPIRY). Expired batches are put on hold automatically every night.
- **Stock alerts**: reorder, stock-out and near-expiry alerts for your stores (also in the bell and by email, one message per store and type).

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11. Labels (*Logistics → Labels*)

Shelf labels (barcode + bin) and batch labels (GS1 DataMatrix with GTIN, expiry and batch where the item has a GTIN). Print from the item, bin or GRN.

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12. Common messages

Message	Meaning / what to do
Insufficient stock / would go negative	Less available than asked; check reservations and holds
Batch expired / on hold	Pick another batch; held stock is handled by Quality

Message	Meaning / what to do
Period locked	The date is on or before the store's closed period; use today's date or ask Finance
Item frozen by a stock take	Finish or cancel the count first
Second signatory required	Controlled drug: a witness with the controlled-drug permission signs
Over-receipt	More than ordered; receive only what was ordered, or ask Procurement to amend the PO
`NON_STOCK_LINE`	Use *Logistics → Service certificates*
`ACCEPTANCE_REQUIRED`	The requester / in-charge must accept the delivery first

Pharmacy (pharmacists and pharmacy technicians)

For pharmacists and pharmacy technicians

For pharmacists and pharmacy technicians in the main pharmacy and the outpatient, inpatient, theatre and emergency outlets. Read Getting started first. For stock work in your pharmacy (receiving, transfers, adjustments, stock takes, issues to wards), see Stores. The rules are the same.

	Pharmacist	Pharmacy technician
Receive / enter prescriptions, dispense	✔	✔
Verify prescriptions, **override** clinical alerts	✔	–
Approve P / POM sales at the till, approve stock adjustments and stock takes	✔	–
Manage the interaction list, catalog, recalls and quality holds	✔	–
Controlled and cold-chain items	✔	✔

Your menu: *Pharmacy & billing* (Pharmacy worklist, Ward supply, Encounters, Patients, Drug interactions, Sales & receipts), plus the *Inventory*, *Wards & issues* and *Planning* screens for your outlets, and *Point of sale* if you sell at the counter.

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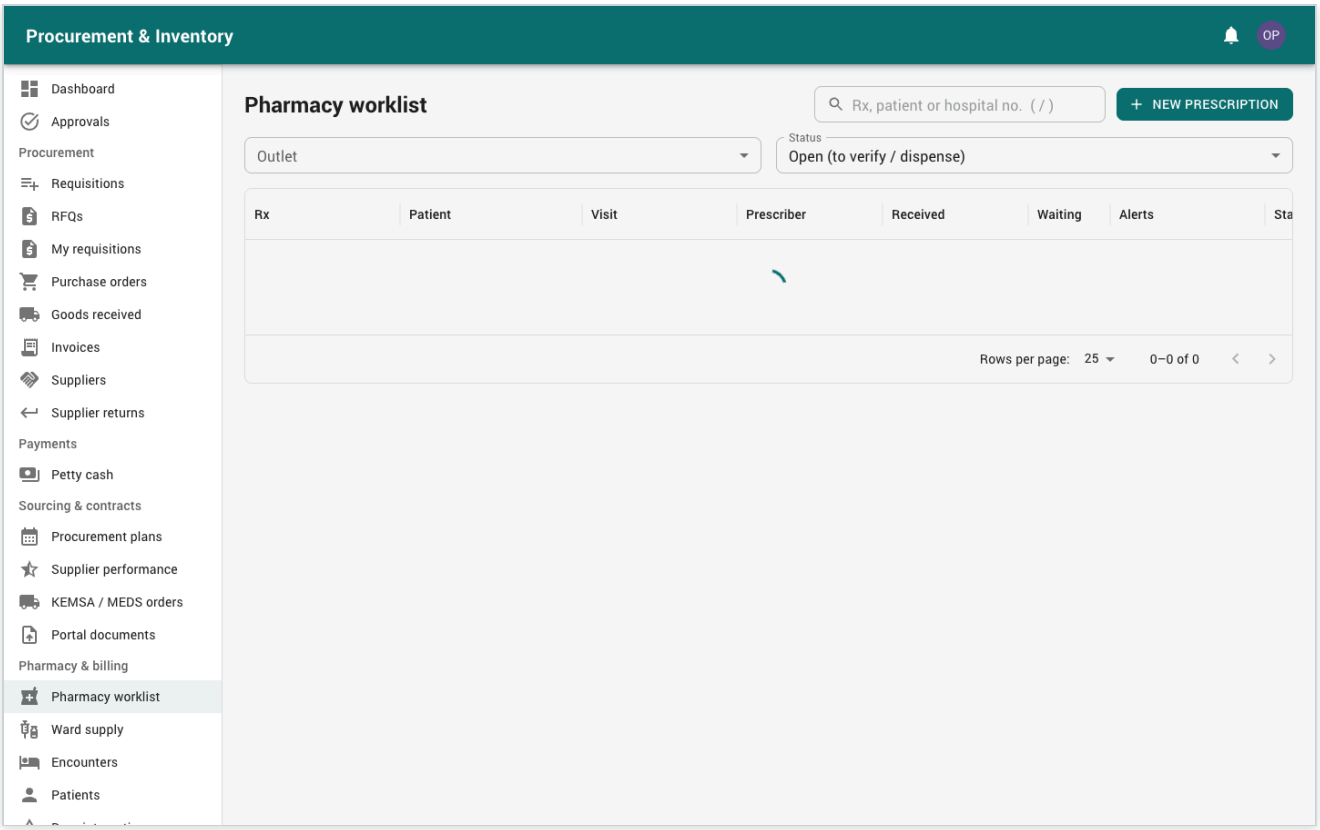
1. Patients (*Pharmacy & billing → Patients*)

The hospital's HMIS is the patient register. This system keeps a copy of what dispensing needs: hospital number, name, date of birth, sex, weight, pregnancy, payer / scheme and **allergies**.

- *Find in HMIS*: search by hospital number or name and import the patient (refreshed automatically when the copy is old).
- *New patient*: only when the HMIS is not connected.
- Keep **allergies**, **weight** and **pregnancy** up to date: the clinical checks use them.

---

2. The worklist (*Pharmacy & billing → Pharmacy worklist*)



The pharmacy worklist

Choose your **outlet**. The worklist shows open prescriptions, oldest first, with waiting time and alert badges. It refreshes every 30 seconds.

Prescriptions arrive:

- **from the HMIS** automatically, as doctors order them (mapped drug codes; orders with an unmapped code wait on the HMIS dashboard for the administrator);
- **entered here**: *New prescription* → patient, encounter / visit, prescriber; *Add medicine* at **generic** level with dose, frequency, duration, route and quantity.

---

3. Clinical checks

Checks run when a prescription arrives, at verification and at every dispense, against the other lines **and the patient's active medications** (dispensed in the last 30 days):

- drug–drug **interactions** (including each component of combination products);
- **allergies** by substance or class ("penicillins" matches a penicillin antibiotic);
- **duplicate therapy** (same substance: moderate; same class: minor);
- **minimum age** and **pregnancy** warnings.

Severity	What happens
Contraindicated, major	**Blocking**: verification and dispensing stop until a pharmacist overrides the alert
Moderate, minor	Shown in colour; you can continue

Overriding (pharmacists only): *Override clinical alert → Clinical reason (logged and reported)*. Overrides can't be edited or deleted. They are listed under *Drug interactions → Alerts raised vs overridden*. *Re-run checks* after correcting the prescription or the patient's details.

The starter interaction list is marked **DEMO**: it is an illustration, not a clinical reference. Your hospital should import a licensed list (*Drug interactions → Import interactions (CSV)*).

---

4. Verify and dispense

1. Open the prescription. Check the clinical checks panel.
2. *Verify* (pharmacist).
3. *Dispense*:
    - choose the **brand** (any brand of the prescribed generic; a substitution is recorded);
    - the **batch** is chosen FEFO; another batch needs the FEFO override and a reason;
    - **partial supply** is allowed. The balance stays open;
    - controlled drugs: a **witness** signs with their username and password.
4. Billing:
    - **billed here** (local billing): the dispense is invoiced. Take payment (cash, M-Pesa, card, insurance / scheme) and print the receipt;
    - **billed in the HMIS**: each item given is charged to the patient's HMIS bill automatically (*Charged to the HMIS bill*); no payment is taken here.
5. The dispense record (batch, expiry, brand) is sent back to the HMIS, so it shows on the patient's record.

Timestamps are kept: received → verified → first dispensed → completed. They feed the turnaround reports.

Cancel undispensed balance (with a reason) closes what will not be supplied.

Returns

Return medicines: choose the dispensed lines and quantities. Returned stock goes *Back into sellable stock (same batch)*. A credit note is raised, or a negative charge on the HMIS bill, and the HMIS record is updated.

---

5. Inpatients: ward supply (*Pharmacy & billing → Ward supply*)

Daily and unit-dose orders for admitted patients, by ward and bed. Each line shows what is needed **today** (daily quantity – already given today). Select lines and dispense the round (one dispense per prescription).

Ward stock top-ups are not done here; they are the ward's internal requisitions, issued under *Wards & issues → Issue vouchers*.

---

6. Sales receipts and eTIMS

Every invoice and credit note is sent to **KRA eTIMS** through the outlet's control unit. Until eTIMS accepts it, the receipt says **"eTIMS PENDING — THIS IS NOT AN OFFICIAL RECEIPT"**. Once accepted you can reprint the signed receipt (marked COPY) from *Sales & receipts*.

Items without an eTIMS mapping can't be invoiced. Tell the administrator or finance (*KRA eTIMS* console).

Invoices can't be changed after issue; corrections are **credit notes**.

---

7. Stock in the pharmacy

- **Receiving**: GRNs with batch and expiry (and fridge temperature for cold-chain items). See Stores, section 3.
- **Transfers** from the main pharmacy to outlets, **adjustments**, **stock takes**: see Stores.
- **Near expiry, reorder, alerts** (*Planning*): see Stores, section 10.
- **Controlled drugs** sit in CD bins; every movement needs a witness.
- **Cold chain**: fridge readings and excursions are under *Quality & safety → Cold chain*. An excursion can quarantine the fridge's batches.

8. Quality and safety (pharmacists)

- **Complaints** (*Quality & safety → Complaints*): report a product quality problem or an adverse drug reaction (PPB report type, reference, date sent); print the report.
- **Recalls** (*Quality & safety → Recalls*): a recall quarantines every affected batch at once. The **trace** lists the stores holding them, wards issued and **patients dispensed (with phone numbers)**, so they can be contacted.
- **Holds**: batches under investigation can't be dispensed or sold until Quality releases them.

9. Selling at the counter

Over-the-counter sales are on the **Till**: see Point of sale. Pharmacists also **approve P / POM sales** for cashiers (your username and password at the till) and can **void and refund**.

10. Reports

Reports → Pharmacy performance: prescriptions and items dispensed (per pharmacist, outlet, day), dispensing turnaround, revenue and margin, top-moving / slow-moving items, ABC and VED analysis, fill rate and stock-outs, alerts overridden, expiry losses.

11. Common messages

Message	Meaning
Blocking clinical alert	Contraindicated / major interaction, allergy or pregnancy flag; a pharmacist must override with a reason
Batch on hold / expired	Pick another batch; Quality manages holds
No stock	Nothing usable at this outlet; request a transfer from the main pharmacy
Item not mapped for eTIMS	The item can't be invoiced until finance / admin maps it
Witness required	Controlled drug: a second, different person with the CD permission signs

Point of sale (cashiers and supervisors)

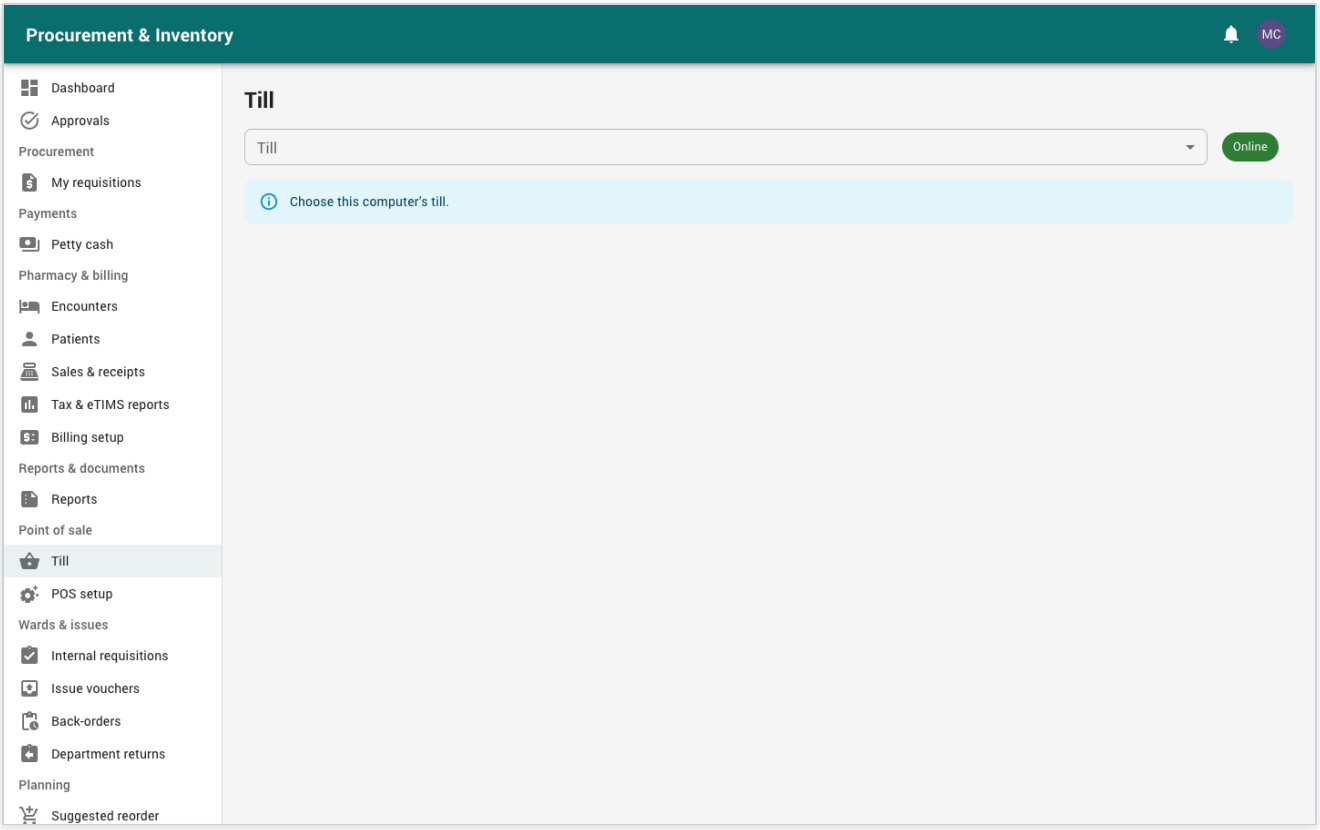
For cashiers and supervisors

For cashiers selling over the counter at a pharmacy till, and for supervisors (pharmacists, finance) who approve sales, overrides and cash-up variances. Read Getting started first.

Your menu: *Point of sale → Till* (cashiers); supervisors also have *Shifts & cash-up*, *POS reports* and *POS setup*.

---

1. Open your shift



The till

1. *Point of sale → Till*, choose your **till**.
2. *Open shift*: count the drawer and enter the *Opening float (cash in the drawer)*.

You can have only one open shift, and a till only one cashier at a time.

2. Selling

1. **Add items**: scan the barcode, or search by brand, generic or code. **Enter** adds the item; **F2** jumps to search; **F9** goes to payment. Quantities are in base units, so you can sell loose tablets from a pack. Stock left and *Low stock* are shown.

2. **Customer** (optional): pick a registered customer for their price list, credit account, loyalty points, or a KRA PIN on the invoice (business buyers).
3. **Discounts and promotions**: promotions apply automatically. You may give a manual discount up to your limit (cashier **5 %**, pharmacist 15 %); above it a supervisor with override rights enters their username and password.

Items that need a pharmacist

Sales class	Rule
OTC	Sell freely
P (pharmacy only)	A pharmacist approves: *Pharmacist approval*, their username and password
POM (prescription only)	Pharmacist approval **and** a *Prescription reference*
CD (controlled)	**Cannot be sold at the till**

Some medicines have a **limit per sale** (e.g. paracetamol) across all brands.

Safety check

The basket is checked for interactions (and against the customer's recent purchases, allergies and pregnancy when they are a registered patient). A blocking interaction needs a pharmacist and a *Reason to proceed despite the interaction*.

3. Taking payment

- **Cash**: enter *Cash tendered*; the change is shown.
- **M-Pesa**:
  - **STK push**: enter the *Customer phone* → *Send prompt*. The customer confirms on their phone and the payment is recorded automatically. If they don't answer in time, try again or choose another method;
  - **Paid to the till / paybill**: pick the customer's payment from the list of confirmations;
  - **Typed code**: enter the M-Pesa code from the customer's SMS. It is accepted but flagged until Safaricom confirms it (your hospital may require the match).
- **Card**: enter the slip number.
- **Split**: combine methods until the total is covered.
- **On account**: for customers with a credit limit.
- **Loyalty points**: 1 point per KES 100 paid; each point is worth KES 1 once the customer has at least 50.

Then *Print receipt* (80 mm) and *Next customer*.

eTIMS. Every sale is a tax invoice sent to KRA eTIMS. The receipt carries the control-unit number and a QR code. If eTIMS hasn't answered within a few seconds, the receipt says **"eTIMS PENDING — not an official receipt"**. The sale is sent as soon as possible and the signed receipt can be reprinted later from *Sales & receipts*.

4. Voids and refunds

- **Void** (current shift only): open the sale → *Void* with a reason. Needs void rights or a supervisor.
- **Refund**: *Refund* the items and quantities, with a reason (refund rights or a pharmacist / supervisor).

Both create a **credit note** and put the stock back on its batch. Cash paid out comes off **your** drawer. M-Pesa refunds are not automatic: refund in cash, or ask finance for a manual M-Pesa payment.

Every void, refund, override and approval is logged.

5. Working offline

If the connection drops, the till **keeps selling** from its saved catalog:

- each sale is queued on the till and the receipt says eTIMS PENDING;
- **no controlled drugs**; P / POM only if you are a pharmacist; M-Pesa only by typed code (STK needs the connection).

When the connection returns, the queue is sent automatically and each sale becomes an invoice **once**. A sale refused at sync (for example, no stock left, or too old) is listed on the till; tell your supervisor. **Don't close the shift until the queue has been sent.**

6. Cash-up and closing

1. *X report* at any time: sales by payment method, voids, refunds, discounts, payments on account, M-Pesa codes not yet confirmed, and the **expected cash**.
2. At the end: *Cash up* → count the drawer by denomination.
3. Within **KES 50** of the expected cash, the shift closes and the **Z report** prints.
4. A bigger variance waits for a **supervisor**, who can *Accept the variance* or *Reject (follow up with the cashier)* under *Point of sale → Shifts & cash-up*. A cashier can never approve their own variance.

7. For supervisors

- **Shifts & cash-up**: open and closed shifts, expected vs counted cash, variances to decide.
- **POS reports**: sales by cashier, shift, item, category, payment method or hour, OTC margin, and the voids / refunds / overrides log (CSV).
- **POS setup** (manager): tills (store and default price list), customers and credit limits, promotions, and the M-Pesa connection (Daraja sandbox / production or mock, till or paybill, keys, callback address).

8. Common messages

Message	What to do
Controlled drug cannot be sold	Refer the customer to the dispensing pharmacy with a prescription
Pharmacist approval required	Call the pharmacist to enter their credentials
Above your discount limit	A supervisor authorises
Limit per sale exceeded	Reduce the quantity
M-Pesa prompt timed out	Send again, or take another method
Variance outside tolerance	Recount; if it is right, the supervisor decides

Finance (finance officers, accountants and the finance manager)

For finance officers, accountants and the finance manager

For the **Finance / Accounts** role (supplier invoices, payment vouchers, releasing payments, M-PESA reconciliation, QuickBooks, petty cash floats) and the **Finance manager** (also approves orders, invoices, payments, petty cash and replenishments, and edits approval workflows). Read Getting started first. The float **custodian**'s daily work is in Petty cash custodian.

Your menu: *Procurement* (Purchase orders, Invoices, Budgets), *Payments* (Payables, Payment vouchers, Petty cash, Petty cash floats, M-PESA reconciliation), *Pharmacy & billing* (Accounts & currencies, QuickBooks, Sales & receipts, Tax & eTIMS reports, KRA eTIMS), *Reports*.

Segregation of duties is enforced:

- whoever captures an invoice, prepares a voucher or raises a request can't approve it;
- the person who **releases** a payment can't be its preparer or one of its approvers;
- every blocked attempt is refused and written to the audit log.

So at least **two finance users** are needed (in the sample data, `finance` and `accountant`), plus the finance manager.

---

1. Supplier invoices (*Procurement → Invoices*)

Capturing

1. Open the **purchase order** → *Capture invoice*. Invoices are always captured against an order.
2. Enter the supplier's invoice number (unique per supplier), invoice date, due date, and for each line the quantity and price invoiced. Attach the invoice scan.
3. *Capture and match*. The **three-way match** compares each line with the order and with what was received (GRN, or the acceptance certificate for services):

Flag	Meaning
Not received	Invoiced, but nothing received or accepted yet
Invoiced > received	More invoiced (including earlier invoices) than received
Invoiced > ordered	More invoiced than ordered
Price variance	Price differs from the order beyond the tolerance (2 %, plus an absolute limit if set)
NOT_ON_ETIMS	The supplier hasn't declared this sale on KRA eTIMS

4. *Re-run match* after a late GRN or acceptance. *Submit for approval*.

Foreign-currency invoices use the order's currency at the invoice-date rate and keep a KES **base total**. **Withholding tax** (from the order lines) is shown with the **net payable**; it is deducted at payment.

Approval

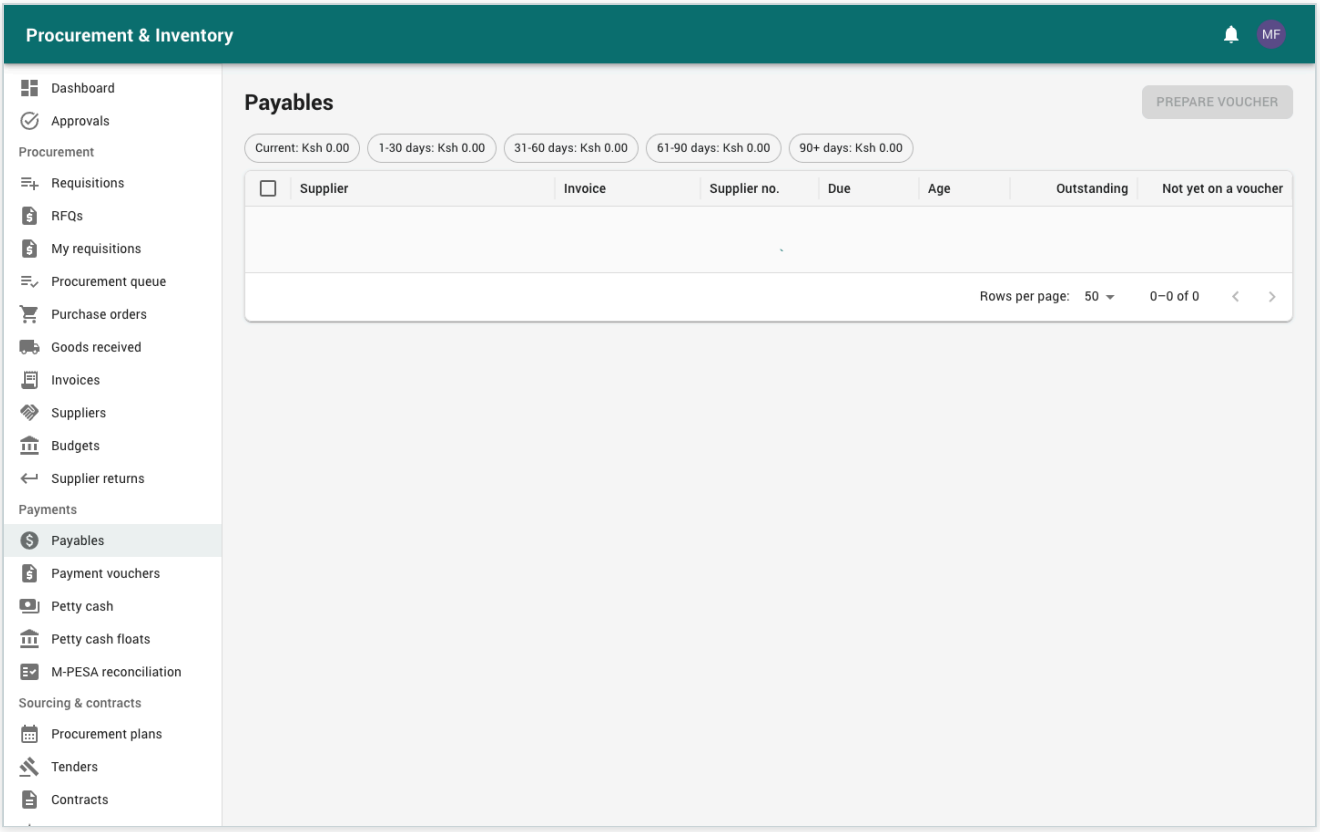
- Exceptions (any variance) go through the approval workflow (Finance).
- Your hospital may auto-approve invoices that match within tolerance.
- Approval books the **actual** against the budget and releases the commitment.

Statuses: Captured → Pending approval → Approved → Scheduled → Partially paid → **Paid**; also Rejected, Returned, Cancelled. When an order is fully received and invoiced, it becomes **Invoiced**; when its invoices are paid, **Paid**.

---

2. Paying suppliers

Payables (*Payments → Payables*)



Payables with the ageing buckets and Prepare voucher

Approved invoices still owed: outstanding amount, amount not yet on a voucher, due date and **ageing** (current, 1–30, 31–60, 61–90, 90+ days).

Preparing a payment voucher (preparer)

1. Tick **one supplier's** invoices in **one currency** → *Prepare voucher*.
2. For each invoice: pay in full or in part. WHT is deducted in proportion. The voucher shows the amount settled, the **WHT** and the **cash paid to the supplier**.
3. *Pay by*:

- bank transfer (EFT), cheque, cash;
- **M-PESA PayBill** (business number + account number);
- **M-PESA Till** (buy goods);
- **M-PESA B2C** (phone number; entered as 07…, 2547… or +2547…).

M-PESA pays KES in **whole shillings** only.

4. *Submit for approval*. The invoices become **Scheduled**. An invoice can't be put on more vouchers than it has left to pay.

Approval (default)

Amount	Approvers
Up to 1,000,000	Finance manager
Above 1,000,000	Finance manager → Hospital management

Releasing (releaser: a different finance user)

1. Open the **approved** voucher.
2. - **Bank / cheque / cash**: *Record payment* with the bank reference or cheque number.
   - **M-PESA**: *Send via M-PESA*.
3. The invoices' paid amounts update (Partially paid / Paid).

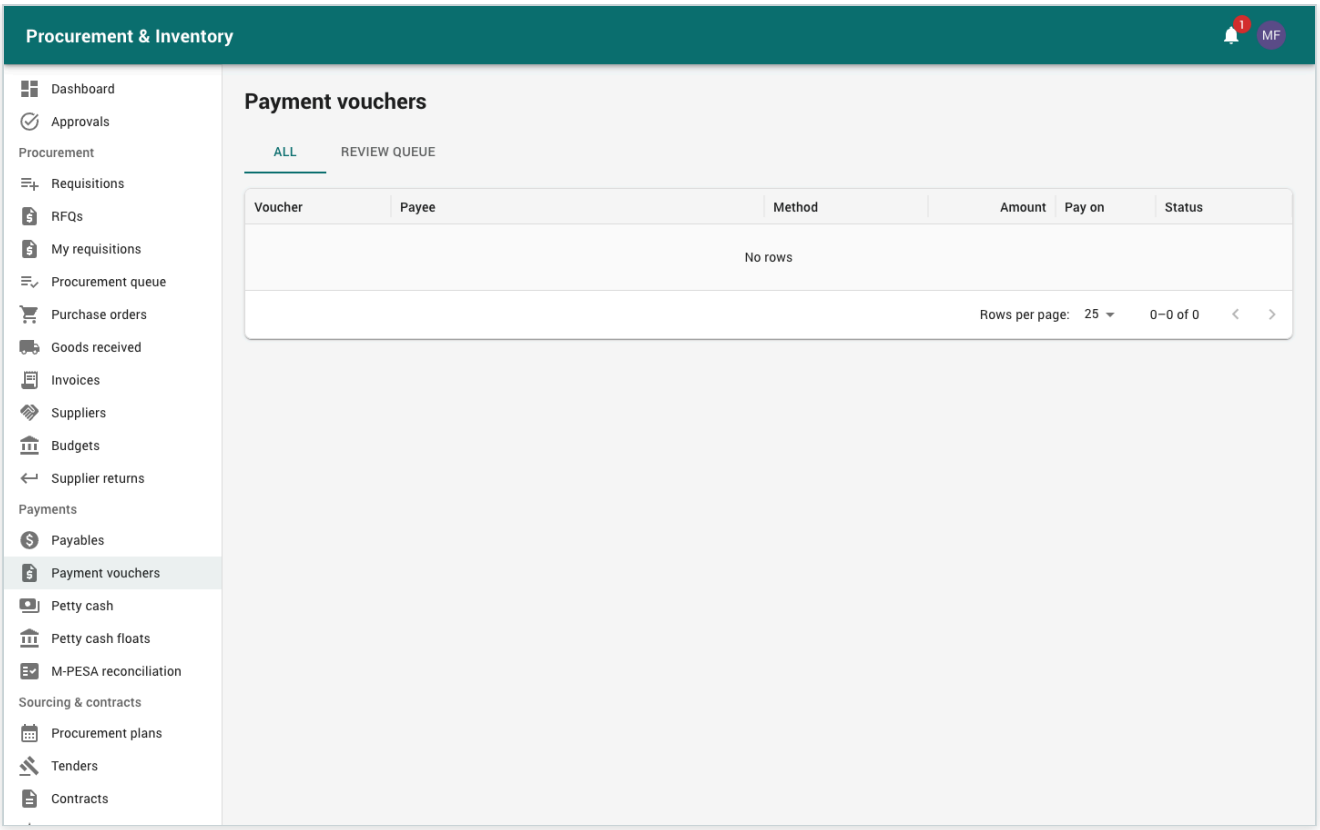
Release is **blocked** until every approval level is complete, and for the preparer and the approvers. *Cancel voucher* is possible before release.

What happens with M-PESA

Result	Voucher
M-PESA confirms	**Paid**, with the M-PESA receipt, recipient name and charges
Refused (wrong number, no funds…)	**Failed**: nothing was sent. Fix the cause and *Send again via M-PESA*
No answer	**Processing**: the system asks M-PESA for the status every few minutes. If it was never processed it is sent again (up to 3 times)
Still unknown after 6 checks	**Review**: nothing more is sent until someone resolves it

A payment is never sent twice. Double clicks and second releases are refused (`ALREADY_DISPATCHED`).

Review queue (*Payments → Payment vouchers → Review queue*, reconcilers)



Payment vouchers, with the review queue

Check the **M-PESA statement** first, then *Resolve*:

- **Paid**: enter the M-PESA receipt; or
- **Not paid**: frees the voucher for a fresh attempt.

Never pay again before you have checked the statement. Every resolution is audited. *Check pending payments now* asks M-PESA about every payment still waiting.

Reversal: *Reverse on M-PESA* (with a reason). When M-PESA confirms, the voucher is Reversed and the invoices are owed again.

The voucher **PDF** prints the invoices, WHT, net amount and approval signatures.

---

3. M-PESA reconciliation (*Payments → M-PESA reconciliation*)

1. Export the organisation **statement CSV** from the M-PESA portal (Receipt No., Completion Time, Details, Paid In, Withdrawn) → *Import* (upload or paste). Re-importing is safe.
2. *Reconcile* a day or a date range (the previous day is reconciled automatically at 06:00).

Exceptions to work through:

Exception	Meaning
`NOT_ON_STATEMENT`	Paid here but not on the statement
`AMOUNT_MISMATCH`	Receipt found, different amount

Exception	Meaning
`NOT_IN_SYSTEM`	A withdrawal on the statement this system didn't make
`STATUS_MISMATCH`	On the statement, but recorded here as failed or unknown: resolve it as paid
`PENDING`	No result yet

Charges are totalled separately. Supplier payments and petty cash are both covered.

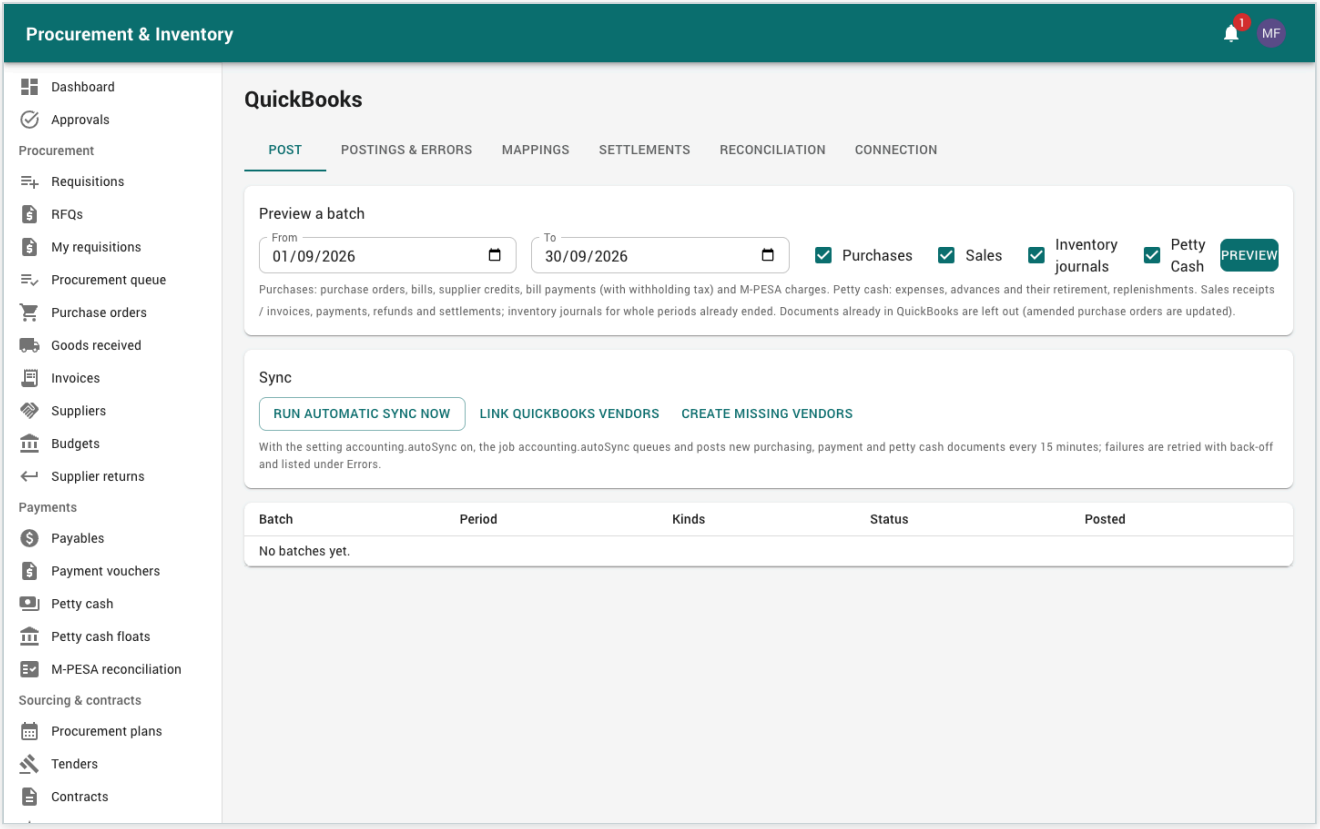
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4. Petty cash (finance side)

- **Floats** (*Payments → Petty cash floats*, `pettycash.manage`): create a float per branch or department with its **custodian**, **float limit**, **per-transaction limit**, **replenishment threshold** (*Replenish below*), cash GL account and whether it *Pays by M-PESA*.
- **Approvals**: Finance checks coding and documents on every petty cash request; the Finance manager approves 10,000–50,000 and above, and **replenishments**.
- **Replenishment**: raised automatically when a float drops below its threshold (or by hand) for the amount up to the limit. After approval, the custodian records the cash received with its reference.
- The custodian's work (paying out, retirements, counts) is in Petty cash custodian.

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5. QuickBooks Online (*Pharmacy & billing → QuickBooks*)



QuickBooks: Post, Postings & errors, Mappings, Settlements, Reconciliation, Connection

Set-up (*Connection*, *Mappings*)

- **Connection**: client ID / secret from the Intuit developer app, sandbox or production, *Connect to QuickBooks*. *Inventory journals from (go-live)*: nothing earlier is posted.
- **Mappings**: map every local key to a QuickBooks account, class or tax code:
    - **control accounts**: AP, AR, GRNI, VAT, stock in transit, adjustments, write-offs, opening balance equity, default bank, **M-PESA clearing**, **WHT payable**, **bank charges**, **staff advances**;
    - per **item category**: inventory, cost of sales, sales income, consumption;
    - per **payment mode / payment method**, per **tax category**;
    - **GL accounts**, and optional **store and department classes**.

  Not mapped counts show the gaps.
- **Vendors**: *Link QuickBooks vendors* matches existing vendors by KRA PIN or name; *Create missing vendors* creates the rest.

What gets posted

Here	In QuickBooks
Approved LPO / LSO	Purchase order (updated when amended)
Approved supplier invoice	Bill (stock lines to GRNI, non-stock lines to their GL account)
Paid voucher	Bill payment (+ vendor credit for WHT)
M-PESA charges	Purchase (bank charges)
Petty cash payment / advance / retirement / replenishment	Purchase / journal entries / transfer
Sales, credit notes, insurer settlements	Sales receipt, invoice + payment, refund / credit memo
Stock movements of a closed month	Inventory journal entry

Posting

1. *Post* → choose the period and kind → *Preview a batch*. Documents missing a mapping are **Blocked**, with what to map. Documents dated on or before the books' closing date are blocked as *Period closed*.
2. Post the **Ready** documents. They go in order (bills before their payments) and are **never posted twice**.
3. With automatic sync turned on, new purchases, payments and petty cash are posted every 15 minutes (*Run automatic sync now* runs it at once).

Errors (*Postings & errors*): QuickBooks unavailable → **Pending**, retried automatically. Rejected → **Failed**: fix the cause (usually a mapping), then *Rebuild and send*.

Each purchase order, invoice, voucher and petty cash voucher shows a **QuickBooks chip**: synced / pending / failed / not sent.

Reconciliation: compares, per account and period, the local documents, what was posted and the QuickBooks movement. A difference means something was entered or changed directly in QuickBooks. **Changes made in QuickBooks do not flow back here.**

Settlements: *Record a settlement* for insurer / corporate payments against their invoices.

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6. Master data

- **Accounts & currencies** (*Pharmacy & billing → Accounts & currencies*): GL accounts (*New account*, or *Import from QuickBooks*), projects, currencies and **exchange rates** (*Save rate*). Foreign documents use the latest rate on or before their date. **Keep rates current**: RFQ comparisons, orders and invoices depend on them.
- **Budgets** (*Procurement → Budgets*): department × vote head × year, amount, and *When exceeded*: *Warn* or *Block approval*.

7. Finance manager

- **Approvals**: orders ≥ 500,000 (after procurement), invoice exceptions, payment vouchers, petty cash 10,000+, replenishments, staff requisitions above 500,000, plan and contract variations.
- **Workflows** (*Approvals → Workflows*): edit approval rules (document type, amount band, department, category, currency) and steps (role or named user, approvals needed, remind / escalate after N hours). Documents already in approval keep the old version.
- **Period locks** for stores and **stock adjustment** approvals.

8. Reports

Reports → Payments: AP ageing · Payments by method · Payment schedule / cash requirement · M-PESA log and reconciliation · Withholding tax and VAT summary. *Petty cash*: cash book, pending, spend, outstanding advances, replenishments. *Procurement*: received not invoiced (accruals), match exceptions, budget vs actual. *Audit*: QuickBooks sync status, payment control exceptions (duplicate payments, split requests, payments with nothing received). *Tax & eTIMS reports*: VAT output and input, credit notes, eTIMS reconciliation.

9. Common messages

Message	Meaning
`USE_PAYMENT_VOUCHER`	Invoices are paid through a voucher, not directly
Release blocked: not fully approved	Wait for every approval level
Release blocked: preparer / approver	Another finance user must release
`ALREADY_DISPATCHED`	The M-PESA payment is already on its way; don't retry
Whole shillings only	M-PESA amounts can't have cents

Message	Meaning
Blocked: not mapped	Add the QuickBooks mapping, then preview again
Period closed	The document is dated in a period closed in QuickBooks

Petty cash custodian

For petty cash float custodians

For the person who holds a petty cash float (in the sample data, the accountant holds `PC-ADMIN`). You need the *disburse petty cash* permission (Finance / Accounts has it) and to be named as the float's custodian. Read Getting started first.

Your menu: *Payments → Petty cash* and *Petty cash floats*.

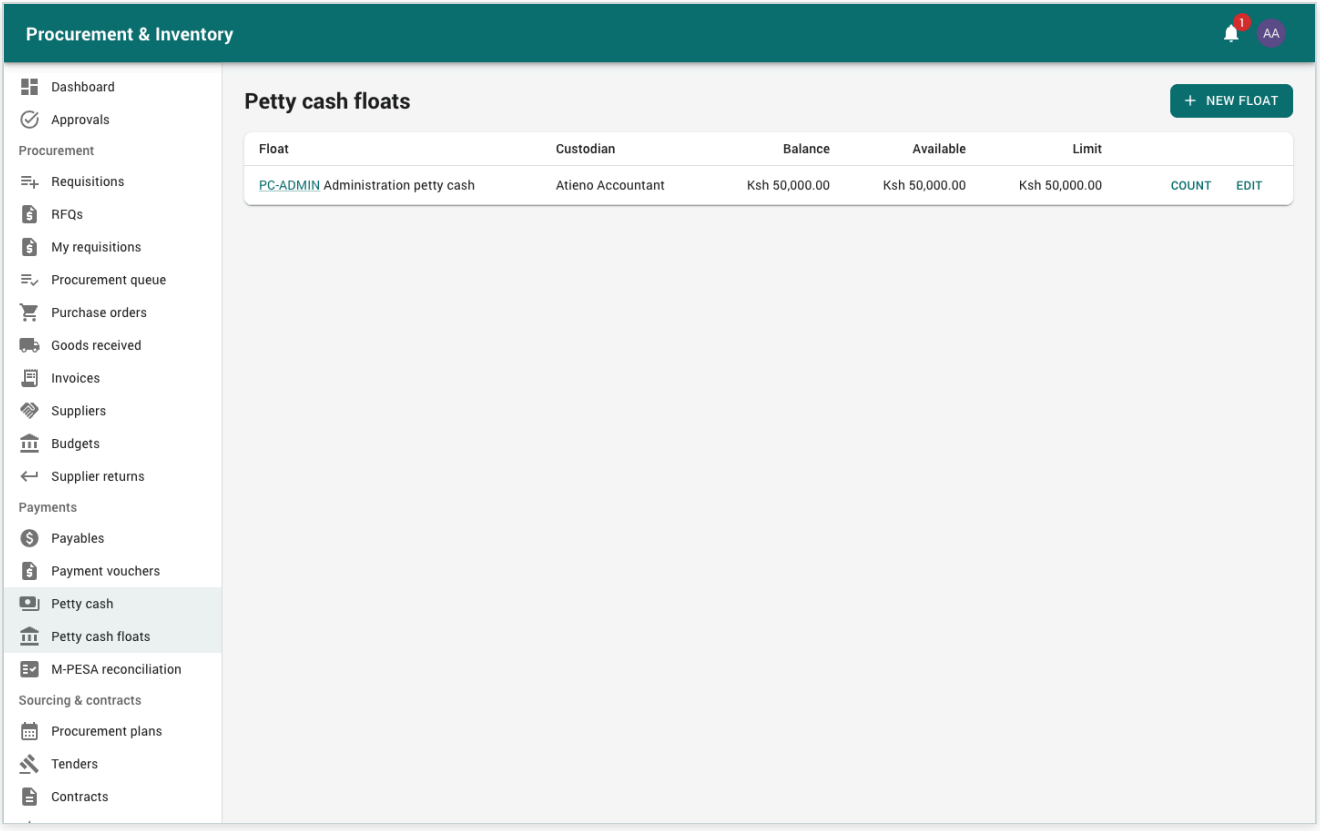
Rules:

- you never approve requests on your own float;
- you only pay requests that are **fully approved**;
- you never pay a request you raised or approved;
- you never check your own retirement.

Attempts that break these rules are refused and audited (`PETTY_CASH_RELEASE_BLOCKED`).

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1. Your float (*Payments → Petty cash floats*)



Petty cash floats

The float page shows the **Balance** (cash in hand per the cash book), **Available** (balance minus requests pending, approved or being paid), the limits, and *Low* when the float is below its replenishment level. The balance changes **only** through the cash book:

Movement	When
Opening	Float set up
Replenishment	Cash received from finance
Disbursement	You paid a request
Refund	Change returned from an advance
Top-up	You paid an advance's overspend
Reversal	A payment reversed

2. Paying approved requests (*Payments → Petty cash*)

The custodian view lists requests on your float that are **approved and waiting for payment**.

Cash

1. Open the request; check the payee and amount.
2. *Pay out cash*; record *Collected by* (who took the cash).
3. The cash book is updated. The payee confirms in the app (*I received it*). The request shows *Receipt confirmed*.

M-PESA (floats that pay by M-PESA)

1. Open the request → *Send via M-PESA*. It goes through the same engine as supplier payments: it is **sent once only**, with status checks and retries.
2. The float moves only when M-PESA confirms (*M-PESA receipt* shown).
3. If the outcome is unknown, the request goes to **M-PESA review**. Check the M-PESA statement, then *Resolve* as **paid** (with the receipt) or **not paid**. Never pay again before checking.

The **petty cash voucher PDF** prints the approvals, who paid and who received.

3. Advances: checking retirements

Staff must retire an advance within **7 days** (daily reminders go to them and to you).

1. Open the advance → *Check the retirement*. Review each receipt: amount, expense account and the attached receipt image.

2. Either **accept**:

    - spent **less**: take back the change; it is recorded as a **Refund** into the float;
    - spent **more**: pay the difference; recorded as a **Top-up**;

    or **return** the retirement to the holder with a comment to correct it.

4. Replenishment

When a payment takes the float below its threshold, a **replenishment** for the amount up to the float limit is raised automatically and goes to the Finance manager for approval (never you). You can also raise one by hand (*Replenish*).

When the cash (or transfer) arrives: open the replenishment → enter *Cash received* and the *Cheque / transfer ref*. The cash book is updated.

5. Cash counts

Count the cash regularly (and always at handover to another custodian):

Petty cash floats → your float → *Cash counts* → *Count* → enter *Cash counted*. The system compares it with the cash book. Any **variance** is recorded and the petty cash managers are told.

6. Cash book / statement

Float → *Cash book* for a period: opening + replenishments + refunds − disbursements − top-ups (± reversals) = closing, with every movement linked to its voucher, and the counts with their variances. Export or print it for month-end.

7. Common messages

Message	Meaning
Not fully approved	Wait for every approval level
You raised / approved this request	Another custodian or finance user must pay it
Only the custodian of the float can pay	You are not this float's custodian
`INSUFFICIENT_FLOAT`	Not enough available cash: request a replenishment
Cannot check your own retirement	Someone else checks it

Supply chain, quality and the tender committee

For the supply chain manager, quality officer and tender committee

Three groups:

- the **Supply Chain Manager**: oversees procurement and stores, approves requisitions, disposals and tender awards, plans supply, manages contracts and the approval matrix;
- the **Quality / Cold-chain Officer**: quarantine and release, cold chain, complaints, recalls and disposals;
- **Tender / evaluation committee members**: score tenders and RFQs.

Read Getting started first. The supply chain manager can also do everything in Procurement and Stores.

Part A: Supply chain manager

1. Approvals

Approvals → My inbox. By default:

- purchase requisitions 100,000 – 1,000,000 (after the procurement manager);
- disposals (all values, before finance);
- tender awards (before management);
- supplier suspension, blacklisting and reinstatement;
- stock adjustments and stock-take variances.

2. The approval matrix (*Approvals → Workflows*)

You maintain the approval rules for every document type: requisitions (REQ), purchase requests (PR), orders (PO), invoices (INV), payment vouchers (PV), petty cash (PCV) and replenishments (PCRP), internal requisitions (SR), plans, contracts, disposals, sanctions…

For each rule:

- **Document**, **amount band** (from / to, KES), optionally *Only for department*, *Only for item category*, *Only for currency*, and a **priority**. The most specific matching rule wins.
- **Steps**, in order:
  - the approver: a role (*Anyone with the role*), a *Role holder in the document's department* (e.g. the requester's head of department), or a named user;
  - *Approvals needed* (e.g. any 2 of the role);
  - *Remind after h* and *Escalate after h* / *Escalate to* (another role).

Editing a rule while documents are in approval keeps the old version for them; the change applies to documents submitted afterwards. Test a change with a small document first.

3. Planning supply (*Planning*)

- **Demand forecasting**: forecasts per item and store (AMC, moving average or Holt-Winters, picked automatically by accuracy), with adjustment factors for outbreaks, new services or programmes. Selected suggestions become a draft PR (or a transfer for satellites). Accuracy (MAPE) is reported monthly.
- **Supply planning**: months of stock and pipeline per item against the minimum and maximum (2 / 4 months), and a month-by-month order schedule.
- **Quantification** per programme: morbidity method next to consumption method.
- **Donor stock**: stock by funding source, with restrictions (no sale, allowed stores and departments) that FEFO respects.

4. Contracts, suppliers and KPIs

- **Contracts** (*Sourcing & contracts → Contracts*): see Procurement § 9.
- **Supplier performance**: monthly scorecards (on time, in full, quality, price compliance, documents). A score below 60 opens a **corrective action** to follow up.
- **Procurement KPIs**: lead times, plan execution, savings, spend by supplier / category / method, spend under contract, OTIF, forecast accuracy.
- **Organisation**: stores, bins, departments; numbering schemes; store **period locks**.
- **Background jobs** (*Administration → Background jobs*): check that the nightly stock, alert and reorder jobs ran; *run now* if needed.

Part B: Quality / cold-chain officer

5. Holds (*Quality & safety → Quarantine & holds*)

Put a batch on hold for inspection, investigation, a complaint, a temperature excursion or a recall. While **any** hold is open, the batch can't be issued, dispensed, sold or transferred in **any** store (write-offs are still allowed).

To decide a hold (someone **other than** the person who placed it):

- *Release for use*, with a decision note; or
- *Reject (raise disposal)*: a draft disposal is created.

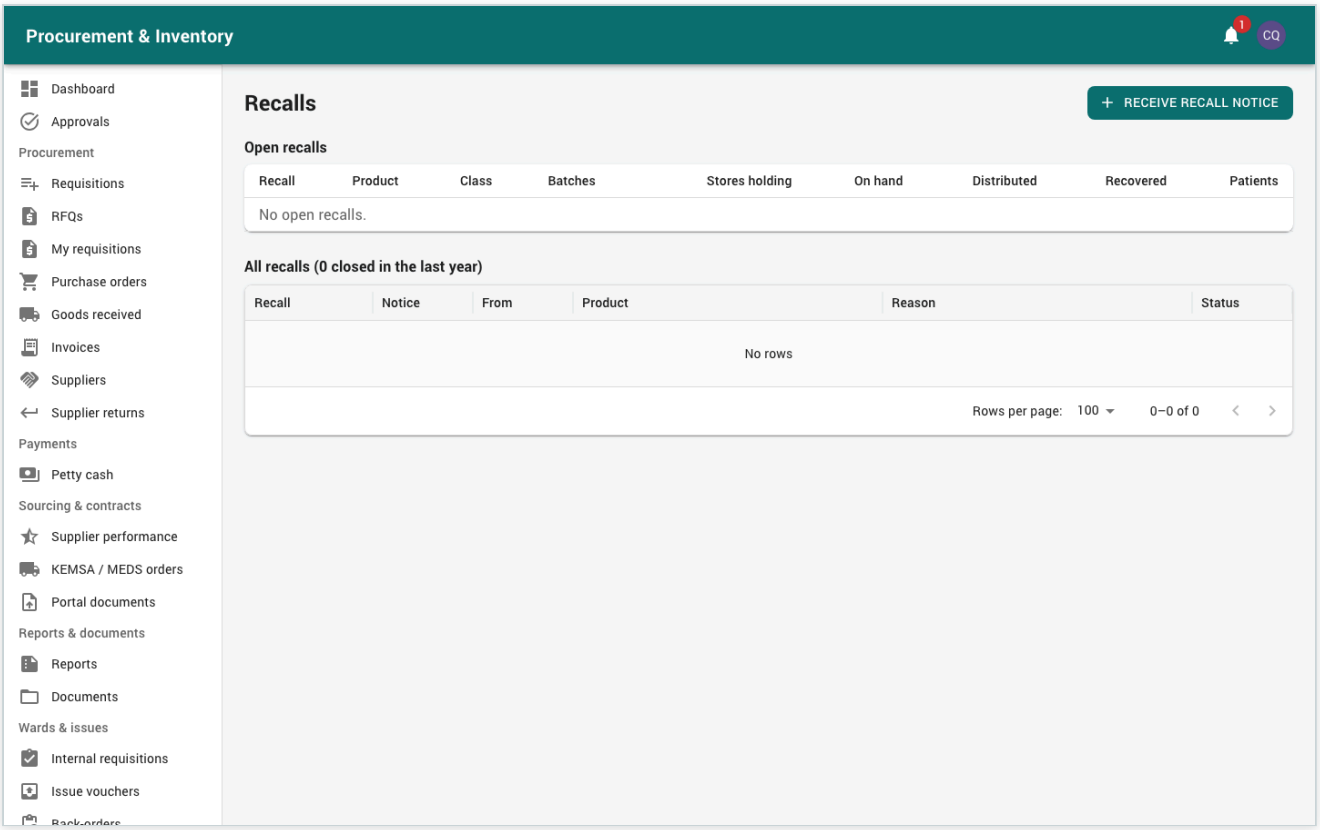
6. Cold chain (*Quality & safety → Cold chain*)

- **Units**: fridges, freezers and cold rooms per store (linked to bins), with ranges: fridge and cold room 2–8 °C, freezer −25 to −15 °C, ambient 15–25 °C.
- **Readings**: enter them by hand, or *Import logger CSV* (timestamp and temperature columns; local Nairobi time).
- An out-of-range reading opens an **excursion**, which (if your hospital turned it on) **quarantines** the unit's batches. Investigate, record the outcome, then release or reject the holds.
- A GRN arrival temperature outside the range also holds the received batches.

7. Complaints (*Quality & safety → Complaints*)

Anyone handling stock or patients can report. You investigate: product, batch, problem, type (**Poor-quality medicine (PQM)** or **Adverse drug reaction (ADR)**), outcome / action taken, optional quarantine of the batch. Record the PPB reference and the date sent, and print the *Printable PPB report*.

8. Recalls (*Quality & safety → Recalls*)



Recalls

1. *New*: notice source (PPB, supplier, manufacturer, internal), notice reference and date, the product and the batch numbers (or all batches), and instructions.
2. Every affected batch in **every store** is quarantined at once; pharmacy and stores staff are alerted.
3. The **trace** lists: on hand per store (quarantined), in transit, issued to wards, **dispensed and sold to patients (with phone numbers)**.
4. Returns from wards or patients: *Put into store (stays quarantined)*.
5. *Close recall*: **rejects** the holds (stock stays unusable until disposed of or returned to the supplier), or **releases** them if the recall was withdrawn.

9. Disposals (*Quality & safety → Disposals*)

For rejected, recalled, expired or obsolete stock:

1. Create the disposal (or it comes from a rejected hold): lines, **method** (destruction, sale, return to supplier, donation, transfer), **disposal committee** (at least 2 members sign off).
2. Approval: up to 100,000 → supply chain → finance; above that → + management.
3. On approval the stock is **written off**.

4. Record completion: place, agent / licensed handler / buyer, waste-transfer note, proceeds of a sale. The **disposal certificate** PDF is generated and stored.

10. GRN inspection

Quality officers can sit on the **inspection committee** for GRNs: *Record results* and *Confirm as committee member* (see Stores § 3).

Part C: Tender and evaluation committee members

11. Tenders (*Sourcing & contracts → Tenders*)

1. You are named on the **evaluation committee** (one member is the chair).
2. Before scoring you must sign the **conflict-of-interest declaration**. If you declare a conflict, you can't score.
3. After the bid **opening**, evaluate in stages:
    - **preliminary**: pass / fail per bidder (decided by majority);
    - **technical**: score each weighted criterion (weights add up to 100); bidders below the pass mark drop out;
    - **financial**: calculated (lowest qualified price ÷ price × 100).
4. The **chair** records the committee's recommendation. At least 3 members must have scored.
5. Once the evaluation is **locked**, scores can't be changed.

12. RFQ evaluations (*Procurement → RFQs*)

When you are on an RFQ's committee: open the RFQ after bidding is closed → *Your technical scores* → score every compliant quotation 0–100 with a comment. The procurement officer completes the evaluation when all members have scored.

13. GRN inspection

Committee members may also be asked to inspect deliveries (*Procurement → Goods received → Record results* / *Confirm as committee member*).

Management and audit

For hospital management and internal audit

For **Hospital Management** (medical director, CEO, administrators: top-level approvals and oversight) and the **Auditor** (internal audit: read-only access to everything). Read Getting started first.

Part A: Hospital management

1. Your approvals (*Approvals* → *My inbox*, or the email link)

By default, management approves the largest and most sensitive documents:

Document	When
Purchase requisitions	≥ 1,000,000 (after procurement and supply chain)
Purchase orders	≥ 5,000,000 (after procurement and finance)
Supplier payment vouchers	> 1,000,000 (after the finance manager)
Petty cash requests	> 50,000 (after HOD, finance, finance manager)
Annual procurement plans	final step
Tender awards	after supply chain
Supplier suspension / blacklisting	after supply chain
Disposals	> 100,000

For each one, check:

- the **document chain** (requisition → quotations → order → delivery → invoice);
- the attachments;
- the **comparative sheet** and **award justification** for awards;
- the budget line;
- the earlier approvers' comments.

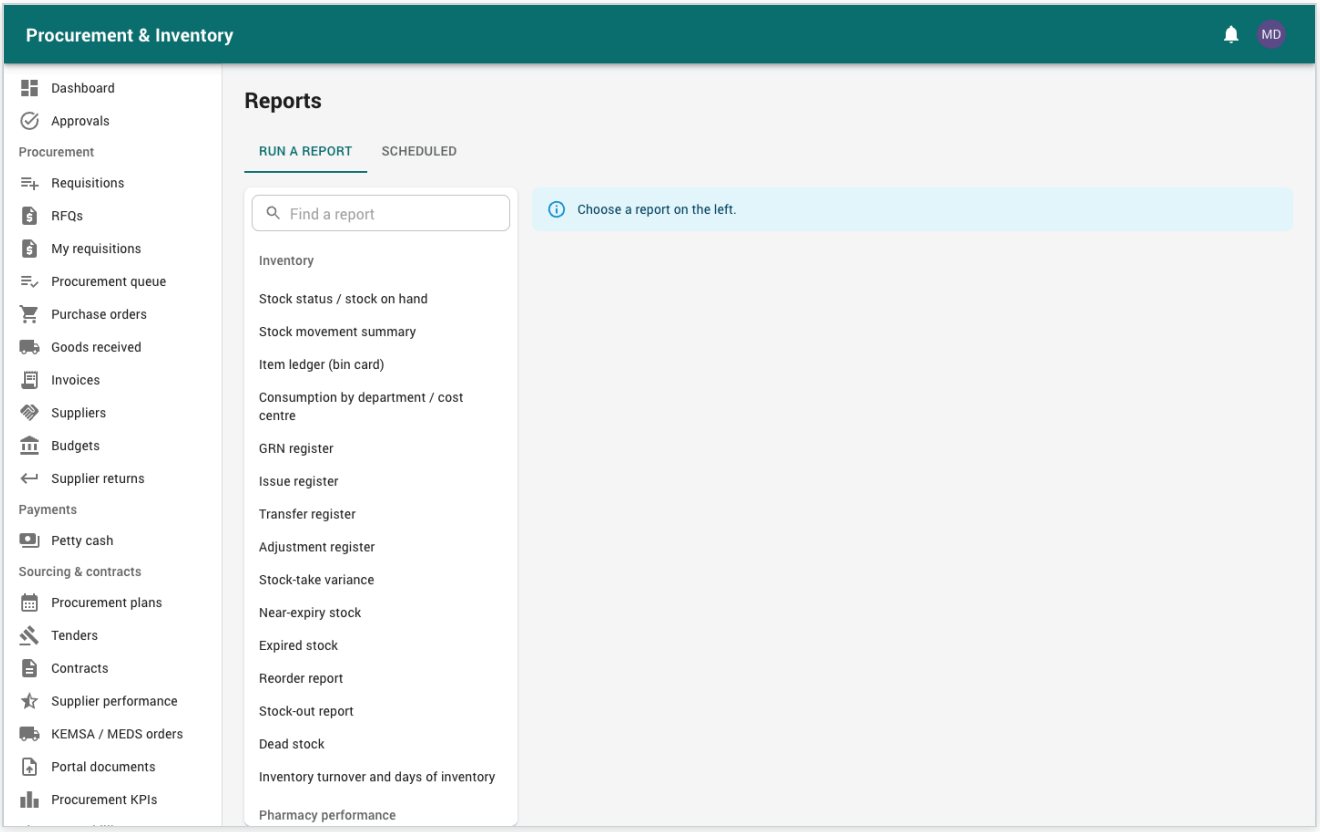
Email approval: the link in the approval email lets you approve, reject or return from your phone without signing in. It works once, only for you, and expires after 72 hours.

Away? Delegate under *Approvals* → *Delegations* (dates, delegate, reason). Your delegate's actions are recorded "on behalf of" you.

2. Dashboard

Tiles: approvals waiting for you, spend this month vs last month, open orders / LPOs (value), invoices to pay, petty cash in hand, overdue advances, M-PESA payments failed or in review, stock value and value expiring, write-offs, integration errors. Click a tile for the detail. Six-month trends: stock value, write-offs, orders, sales.

3. Reports (*Reports*)



The report catalogue

Useful for management, all with CSV / Excel / PDF export and *Email on a schedule* (e.g. a monthly pack):

- **Purchases over time** (by department, supplier, category or GL, with previous-period change and a chart);
- budget vs committed vs actual vs paid;
- procurement cycle time, pending requisitions and purchase requests;
- AP ageing, payment schedule / cash requirement;
- petty cash spend;
- supplier performance and procurement KPIs;
- stock value, expiry losses and write-offs;
- pharmacy revenue and margin.

Part B: Auditor

You can **see** everything (all stores and departments) and change nothing.

4. Audit log (*Administration → Audit log*)

Every create, change and delete, with **before / after values**, who, when, and from which IP address. Secrets and bank details are redacted. The log can't be changed or deleted (the database refuses).

Notable security events:

Event	Meaning
`LOGIN`, `LOGIN_FAILED` (bad password, bad 2FA code, locked)	Sign-ins
`PAYMENT_RELEASE_BLOCKED`	Someone tried to release a payment without full approval, or as its preparer / approver
`PETTY_CASH_RELEASE_BLOCKED`	The same for petty cash
`PAYMENT_RESOLVED`	An M-PESA payment in review was resolved by hand
`MPESA_RESULT_CONFLICT`	M-PESA reported "paid" for an attempt recorded as failed
`TWO_FACTOR_ENABLED / DISABLED`, `PASSWORD_CHANGE`	Account security changes

5. Audit reports (*Reports → Audit*)

- **Approval audit trail**: every approval action in a period with document, step, decision, who acted and **on behalf of whom** (delegations), channel (app or email link) and IP.
- **Payment control exceptions**:
  - **duplicate payments**: an invoice settled more than once, or the same petty cash payee and amount within the duplicate window;
  - **possible splits**: several requests by one requester (or orders to one supplier) within 3 days, each under an approval threshold and together over it;
  - **payments with nothing received or accepted** on the order lines invoiced.
- **QuickBooks sync status and errors**.

Also useful: three-way match exceptions, received-not-invoiced, M-PESA log and reconciliation, WHT / VAT summary, petty cash book and cash-count variances, stock-take variances, adjustments and write-offs registers, alerts overridden in the pharmacy, POS voids / refunds / overrides.

6. Controls built into the system (what you can rely on)

- Nobody approves their own request; each approval step is by a different person; the payment preparer can't approve or release it.
- No payment (supplier or petty cash) can be released before **every** approval level.
- An M-PESA payment is never sent twice for the same voucher (enforced in the database).
- Non-stock purchases never change stock; every stock change is in an append-only ledger.
- Approval-email links are single use, personal and expire.
- Store-scoped users see only their stores.
- Every route needs a permission (checked by an automated test).

7. Documents (*Reports & documents → Documents*)

Search every attached file (delivery notes, invoices, licences, bids, certificates, evidence) by name, record number, note or checksum. Earlier versions are kept, and files are checked against their SHA-256 on download.

System administrator

For the system administrator

For the IT / system administrator who runs the application day to day: users and roles, reference data, integrations and background jobs. Installation, TLS, environment variables, backups and monitoring are in the technical **Deployment guide**. Read Getting started first.

The System Admin role can see and do everything. Use it for administration only. For daily work, give yourself the role you actually work in, and **turn on two-factor authentication**.

---

1. The first administrator

A new production installation has no users. After the reference data is loaded, create the first administrator from the server:

```
node dist/database/seeds/create-admin.js jdoe "Jane Doe" jdoe@hospital.org
```

The one-time password is printed once. Sign in, change it, and turn on 2FA (*My account*). Then create the other users in the application.

2. Users (*Administration → Users*)

New: **username** (lower case), **full name**, email (for notifications and approval emails), phone, **department** (it decides which department approvals the user receives as in-charge, and their default cost centre), **roles**, **stores** (for store-scoped roles) and an *Initial password*.

- The user must change the initial password at first sign-in.
- **Reset a password**: *New temporary password*. The user changes it at next sign-in.
- **Unlock**: accounts unlock by themselves after 15 minutes; resetting the password also helps.
- **Leavers**: untick **Active**. They are signed out at once and keep their history. Never delete or reuse accounts.
- **Store scope**: storekeepers, pharmacists and cashiers see only the stores you assign. Roles with *all stores* (procurement, finance, supply chain, audit, management) see every store.
- **Department in-charges** need the Department / Ward In-charge role **and** the right department. Approvals go to the in-charge of the requester's department.

3. Roles (*Administration → Roles*)

The built-in roles cover the usual jobs (see the index). You can create new roles and tick permissions. Each permission's description says what it allows.

Guidelines:

- Keep segregation of duties. Don't give one person `payments.prepare`, approval of vouchers (through the Finance manager role) **and** `payments.release`. The system blocks the same person acting twice on one voucher, but a role design that relies on it is fragile.
- Give `inventory.handle_controlled` only to staff allowed to handle controlled drugs.
- Supplier users get only the **Supplier** role and must be linked to their supplier.

4. Organisation and catalog

- **Stores** and **Bins** (*Organisation*): main store, pharmacies, satellites; parent store (who supplies whom); quarantine, controlled-drug and fridge bins.
- **Departments** (*Organisation*): code, name and **cost centre**.
- **Items / Generics** (*Catalog*): code, generic, brand, base unit and pack conversions, category, tax category, batch tracking, controlled / cold-chain flags, GTIN / barcode, sales class (OTC / P / POM / CD), VED class.
- **Bulk import** (*Catalog → Bulk import*): CSV / Excel. Validate first, review errors and duplicates, then import.

5. Settings you may be asked to change

Business settings are in the database (ask the vendor or a developer where there is no screen). Common ones, with their defaults:

Setting	Default	Effect
`security.maxFailedLogins` / `security.lockoutMinutes`	5 / 15	Lockout after wrong passwords or 2FA codes
`approvals.linkExpiryHours`	72	Lifetime of approve-by-email links
`requisitions.requireGlAccount`	false	GL account required on requisition lines
`purchasing.rfqMinSuppliers`	3	Minimum suppliers on an RFQ
`purchasing.matchPriceTolerancePct`	2	3-way match price tolerance
`purchasing.autoSendOnApproval` / `autoApproveMatchedInvoices`	false	Automation switches
`purchasing.lpoValidityDays`	60	Order validity
`purchasing.whtRates`	services 5 %	Withholding tax
`pettyCash.duplicateWindowDays` / `retirementDays`	7 / 7	Petty cash checks
`pos.discountLimits`, `pos.varianceTolerance`	5 % cashier / 50	Till rules
`accounting.autoSync`	false	Post to QuickBooks automatically

Setting	Default	Effect
`jobs.schedules`	see below	Cron times of background jobs (applied when the worker restarts)

Approval **workflows** are edited in the application (*Approvals → Workflows*). See Supply chain § 2.

6. Background jobs (*Administration → Background jobs*)

Lists every job, its schedule, last run and result, with *run now*. Key jobs:

Job	When	Why it matters
`payments.mpesaFollowUp`	every 5 min	Follows up M-PESA payments with no result
`approvals.escalate`	every 15 min	Reminders and escalations
`accounting.dispatch` / `autoSync` / `refresh`	5 / 15 min / hourly	QuickBooks
`stock.expiry`, `stock.reorder`, `stock.amc`, `inventory.reconcile`	nightly / hourly	Stock alerts and checks
`purchasing.expireOrders`	01:30	Expires old LPOs
`pettyCash.retirementReminders`	07:00	Chases overdue advances
`reports.scheduled`	every 15 min	Emails scheduled reports
`etims.*`, `hmis.*`	every minute–daily	Integrations

A failed job shows its error. Most are safe to *run now* again.

7. Integrations

- **KRA eTIMS** (*Pharmacy & billing → KRA eTIMS*): one control unit per selling outlet (initialise it), code-table sync, **mappings** (items without a mapping can't be invoiced), the **outbox** (status, attempts, the call log, reprocess failed), and the *Specification notes*.
- **HMIS** (*Pharmacy & billing → HMIS integration*): connection (FHIR, REST, HL7 v2 or mock), *Test connection*, who bills (here or the HMIS), **code mappings** (drug codes, locations, wards, payers). The **dashboard** shows failed messages and **unmapped drug codes**: map them, then reprocess.
- **QuickBooks**: see Finance § 5.
- **M-PESA**:
  - till payments (STK / C2B): *Point of sale → POS setup → M-Pesa*;
  - supplier and petty cash **payouts**: configured on the server only (see the deployment guide). Credentials are never entered in the application.
- **API clients** (for the HMIS pushing invoices or orders): created through the API (`/api/v1/api-clients`, administrator only; there is no screen yet) with their scopes. The key is shown **once** and stored hashed.

8. Monitoring and support

- **Dashboard** integration tiles (eTIMS, HMIS, QuickBooks errors).
- **Audit log** (*Administration → Audit log*): who changed what and when.
- `ops/monitoring/check.sh` on the server (API, worker, jobs, mail queue, M-PESA, QuickBooks, eTIMS / HMIS, backup age). See the deployment guide.
- **Mail not arriving?** The outbox retries; check the SMTP settings with IT. Users can also have switched email off in *My account → Notification preferences*.

Common support requests:

Request	Answer
"I can't see menu X"	Add the role that has the permission; store-scoped roles also need the store
"I'm not getting approvals"	Check the user's role **and department** (in-charge approvals follow the requester's department), and their notification preferences
"Approval went to the wrong person"	Check *Approvals → Workflows*: band, department, category, priority
"Locked out"	Wait 15 minutes, or give a new temporary password
"Lost my phone (2FA)"	An administrator can't read the secret. Give a new temporary password; if 2FA must be removed, a developer clears it, and the user then sets it up again
"eTIMS PENDING on receipts"	Check the eTIMS outbox and control unit; documents are re-sent automatically

9. Backups

Nightly backups and a weekly restore drill are run by IT with `ops/backup/`. See the Deployment guide § 8. Keep the encryption keys and the backup passphrase in the secret store, not with the backups.

Supplier portal

For suppliers' staff

For suppliers' staff who have been given a portal account by the hospital. You see **only your own company's** quotations, tenders, orders and invoices.

1. Signing in

Open the address the hospital gave you and sign in with your username and password. Change the initial password when asked, and turn on two-factor authentication under *My account*. After 5 wrong passwords the account locks for 15 minutes.

2. Supplier home

A summary of open requests for quotation, tenders, orders to deliver and invoices. *Our details with the hospital* shows what the hospital holds about you (KRA PIN, prequalification and its expiry, compliance documents, the bank account masked). Ask the procurement office to update anything that is wrong. Expired prequalification means you can't be invited or ordered from.

3. Requests for quotation (*Requests for quotation*)

1. Open an RFQ you were invited to. Check the items or services, quantities, specification, delivery address and the **closing date**.
2. Enter your unit price for each line, *Delivery (days)*, *Payment terms*, *Valid until*, and any notes. Attach your quotation document.
3. *Send* before the closing date. You can't change it after bidding closes.

The hospital compares quotations on price, compliance and delivery (and may score quality). You'll receive an order if you are awarded.

4. Tenders (*Tenders*)

1. Open tenders, and those you are invited to, are listed. *Request bid documents* to download the tender documents.
2. *Ask a question* during the clarification period. *Clarifications and addenda* are sent to everyone holding the documents. Read them.
3. *Submit a bid*:
    - prices per *Tender lines*;
    - *Technical proposal* and compliance documents;
    - *Bid security* amount and reference;
    - *Bid validity*.

Submit bid before the closing time. Late bids are refused.

4. Bids are **sealed**: nobody at the hospital sees prices before the formal opening. *Withdraw* is possible until the closing time.
5. You'll get an award or regret letter by email.

5. Purchase orders (*Purchase orders*)

1. New orders appear here (and arrive by email as a PDF).
2. *Acknowledge order*, with your reference and the expected delivery date (or *Reject*, with a reason).
3. *Update delivery status*: processing → dispatched → in transit.
4. When you deliver, *Send delivery note or invoice*: number / reference, date, amount, and attach the scanned delivery note or invoice.

Deliver what the order says. Over-deliveries are refused at the store, and goods rejected at inspection are returned with a return note.

6. Invoices and payments (*Invoices & payments*)

Your invoices with their status:

- **Captured / Pending approval**: being checked against the order and the delivery;
- **Approved**: due for payment on the due date;
- **Scheduled**: on a payment voucher;
- **Partially paid / Paid**: with the payment date and reference.

Payment is by bank transfer, cheque or M-PESA (PayBill / Till). **Withholding tax** on services (5 %) is deducted and declared to KRA; the payment shows the amount settled, the WHT and the net paid.

Invoices are matched to what the hospital **ordered and received**. If quantities or prices differ from the order, payment waits until the difference is resolved. Make sure your sale is declared on **KRA eTIMS**: invoices not found on eTIMS are flagged.

7. Help

Contact the procurement office for orders and RFQs, and the finance office for payments.